Rough-In Redlines
User Manual

PDF markup for the trades · Version 2.1

Windows | iPad & iPhone | Android tablets | Rough-In Crew

Step-by-step help for foremen, project managers and office staff.

Open a drawing set, mark it up, count and measure, compare revisions, and send it back to the office: editable in Bluebeam and Acrobat, or flattened for the record. Then share it all with your crew.

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1 Welcome

Rough-In Redlines is a PDF markup app made for electricians and the other trades. Open the drawing set you were sent, mark it up the way you would with a red pen, count and measure, compare a new revision with the old one, and send it back to the office. Nothing has to be set up first.

1.1 What you can do

Redline drawings. Pen, highlighter, clouds, boxes, arrows, callouts, text notes, white-out, stamps such as AS-BUILT, revision triangles and your signature.

Place trade symbols. Over 600 symbols in 18 libraries (receptacles, fixtures, fire alarm, low voltage, plumbing, HVAC and more), your own palettes, and symbols you capture off the drawing.

Count. Click each device, or let Find & Count find every symbol like the one you box. Totals go on a count page and into a spreadsheet.

Measure. The app reads the scale printed on the sheet. Measure distances, runs and areas, or calibrate from a known dimension.

Compare revisions. See what changed between two prints, cloud the changes and bring your markups across.

Send it your way. A PDF that stays editable in Bluebeam and Acrobat, a flattened PDF, a counts CSV, a print or plot, or a **.redline** file that opens fully editable on another PC or iPad.

Fill forms and redact. Fill in fillable PDF forms and sign them. Remove names, prices or phone numbers for good before a drawing goes out.

Work as a crew. With Rough-In Crew, drawings and markups are shared with everyone on the job and stay ready offline.

1.2 The apps and where they run

There is one Rough-In Redlines app per kind of device. They all open the same files and they work the same way; the way you touch them is different (mouse and keyboard, Apple Pencil, or a pen and finger).

Device	App	Notes
WINDOWS	**Rough-In Redlines Preview** (Version 2)	Windows 10 (version 1809 or later) or Windows 11, 64-bit Intel/AMD (x64) or ARM64. Website download or Microsoft Store. Version 1 is called Rough-In Redlines; Version 2 installs next to it as Rough-In Redlines Preview while V2 is being tested.
IPAD	**RR Preview**	iPad and iPhone, iPadOS / iOS 17 or later. Also runs on Apple-silicon Macs.
ANDROID	**Rough-In Redlines** 1.0	Android tablets with Android 11 or later (made for Samsung Galaxy Tab S tablets with the S Pen). From Google Play. Version 1.0 has no Crew.
CREW	**Rough-In Crew**	Built into the Version 2 apps on Windows and iPad. Sign in with your work email.

1.3 How to use this manual

Chapters 2 and 3 get you started and show you around the screen. Chapters 4 to 10 cover each job you do with a drawing: marking up, symbols, counting, measuring, forms, managing sheets and sending your work. Chapter 11 is about working as a crew. When something goes wrong, see chapter 12. Keyboard shortcuts and touch gestures are in chapter 13, and the words the app uses are explained in chapter 14.

Words in **bold** are what you see on the screen: a button, a menu item or a setting. **File → Export → Flattened PDF…** means: open the File menu, then Export, then click Flattened PDF…. Keys look like `Ctrl`+`S`.

When something works differently on one kind of device, a small label shows which one:

WINDOWS the Windows app IPAD the iPad and iPhone app ANDROID the Android tablet app

CREW needs a Rough-In Crew sign-in V2 new in Version 2 PRO about Rough-In Pro

No label means it works the same everywhere.

> **A NOTE ON THE PICTURES**
>
> The Windows pictures in this manual were taken from the Windows app's own screens. Your screen may look a little different: Windows uses its own font, the colors follow your theme (light or dark) and your accent color, and the buttons you see depend on Settings → Tools. All the drawings shown are the built-in sample drawing or made-up test data.

2 Getting started

Install the app, open your first drawing and try the sample. You don't need an account to mark up drawings: only Rough-In Crew asks you to sign in.

2.1 Install on Windows WINDOWS

There are two editions with the same features and the same license key:

- **Website download**: a standard Windows installer (**…_x64-setup.exe**, or **…_arm64-setup.exe** for ARM PCs). Settings → About shows **Edition: Website download**, and **Help → Check for updates…** looks for a newer version.
- **Microsoft Store**: installed and kept up to date by the Store. Settings → About shows **Edition: Microsoft Store**.

You need Windows 10 version 1809 or later, or Windows 11, on a 64-bit PC (x64 or ARM64). 4 GB of memory is recommended for big drawing sets. The app uses the Microsoft Edge WebView2 Runtime, which is built into Windows 11 and current Windows 10.

> **"WINDOWS PROTECTED YOUR PC"**
> If Windows SmartScreen stops the website installer, click **More info**, then **Run anyway**.

Version 2 Preview. While Version 2 is being tested it is called **Rough-In Redlines Preview** (**RR Preview**). It installs next to Version 1 with its own drawings, and keeps itself up to date: it looks for an update when it starts and every few hours, and asks before it installs one (**Help → Check for updates…**).

The first time it starts, the app asks **How will you use Rough-In?** Pick **Trades** (every tool: symbols, counts, stamps, measuring and compare) or **Simple markup** (pen, highlighter, text, shapes, white-out and photos; PDFs out), or click **Skip**. You can change this any time in **Settings → Tools** (see 4.8).

2.2 Install on the iPad and iPhone IPAD

The iPad app comes from Apple. While Version 2 is being tested it is handed out through **TestFlight** as **RR Preview**: install TestFlight from the App Store, open the invite email on the iPad, tap **View in TestFlight**, then **Install**. In TestFlight, turn on **Automatic Updates**. The finished app is called **Rough-In Redlines** on the App Store (**Rough-In** under its icon).

To send feedback from a TestFlight build, take a screenshot, tap the preview, then **Share Beta Feedback** (or open TestFlight → the app → **Send Beta Feedback**).

The first time it opens, the app asks **How will you use Rough-In?** (Trades, Simple markup or Skip), the same as on Windows.

2.3 Install on an Android tablet ANDROID

Install **Rough-In Redlines** from **Google Play**. It needs Android 11 or newer on a 64-bit (arm64) tablet; it is made for tablets with a pen such as the Samsung Galaxy Tab S series with the S Pen. Google Play keeps it up to date.

The Android app is the Windows app with a touch screen layout: everything in this manual about Windows applies, with these changes (more in 4.6):

- There is no menu bar. The ⋯ button at the top right holds the menu items, and a long-press does what a right-click does on Windows.
- The words on screen say **tap**, **long-press** and **this tablet**, and keyboard shortcuts aren't shown (a keyboard cover still uses them).
- Exports ask **Share…** (email, Drive, Teams…) or **Save to…** (a folder).
- Version 1.0 has no Crew, no Word / Excel conversion, and panel schedules are read-only.

2.4 Free, and Rough-In Pro PRO

Every tool is free on up to **3 drawings** at a time. A PDF set is one drawing however many sheets it has; the sample drawing doesn't count; removing a drawing from the home screen frees its place; opening a drawing that's already there never uses another. When you've used your 3, the drawings you have keep working: they still open, mark up and export. Only adding a fourth asks you to unlock Pro.

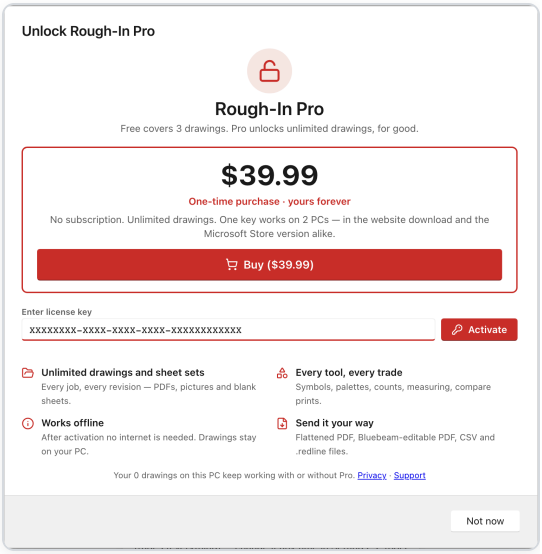
Rough-In Pro is a one-time purchase (**$39.99** in the US) that unlocks unlimited drawings. Pro itself is not a subscription; a Rough-In Crew subscription or license also includes Pro (3.2). Each kind of device has its own store, so Pro is bought separately on Windows, on the iPad and on Android.

	How to buy	Moving to a new device
WINDOWS	A license key, by email. One key works on 2 PCs, in the website and Microsoft Store editions alike.	On the old PC: Settings → Rough-In Pro → **Deactivate this PC**. Then enter the key on the new one.
IPAD	In-app purchase through the App Store.	**Restore Purchases** on any iPad or iPhone signed in to the same Apple Account.
ANDROID	In-app purchase through Google Play.	It comes back by itself on a tablet signed in to the same Google account (or tap **Restore purchase**).

The price shown in the app is the store's price; the App Store and Google Play show it in your own currency.

Unlock Pro on Windows

1. Click the chip at the bottom right of the window (**Free · 2 of 3 drawings — Unlock Pro**), or **Settings** → **Rough-In Pro** → **Unlock Rough-In Pro**. The **Unlock Rough-In Pro** window opens.
2. Click **Buy ($39.99)**. The store opens in your web browser. Finish the purchase there; your license key arrives by email.
3. Copy the key from the email, paste it under **Enter license key** and click **Activate**. You'll see *Rough-In Pro is unlocked on this PC. Thank you!*



The Unlock Rough-In Pro window.

After activation the app works offline. It checks the key now and then when you're online; if it can't reach the license server, Pro keeps working. If the key hasn't been checked for a long time, Settings asks you to connect and click **Check now**.

Settings → Rough-In Pro shows the **Status**, how many **Drawings** you have, the key (only its last characters), who it's licensed to and when it was last checked. **Deactivate this PC** frees one of the key's 2 activations; your drawings stay.

MESSAGES YOU MIGHT SEE

- *That license key wasn't found*: copy and paste it from the purchase email, don't type it.
- *This key is already active on the maximum number of PCs*: on the PC you no longer use, open Settings and click Deactivate this PC, then try again.
- *Couldn't reach the license server*: check the internet connection. Pro keeps working in the meantime.

Unlock Pro on the iPad IPAD

When you add a fourth drawing, the Rough-In Pro screen opens (or go to Settings → **Rough-In Pro** → **Unlock Rough-In Pro**). Tap **Unlock Pro for …** and confirm with the App Store. Bought it before, on this or another device? Tap **Restore Purchases**. If your family uses Ask to Buy, Pro unlocks as soon as the purchase is approved.

The Version 2 Preview builds (RR Preview) have every feature unlocked while they are being tested.

Unlock Pro on Android ANDROID

Tap the chip on the home screen (**Free · 0 of 3 drawings — Unlock Pro**) or Settings → **Rough-In Pro** → **Unlock Pro ($39.99)**, then **Buy ($39.99)** and finish the purchase with Google Play. The app checks for your purchase every time it starts; **Restore purchase** checks right away. A bought Pro keeps working offline.

2.5 Open a drawing

Each drawing you open is **copied into the app**, so the original file is never changed. Markups are saved as you go; there is no Save button to remember.

On Windows WINDOWS

- Click **Open file** on the Home tab (or **File** → **Open…**, `Ctrl` + `O`) and pick a PDF, a picture (JPG, PNG or iPhone HEIC) or a **.redline** file.
- Or drag files from File Explorer anywhere onto the window.
- In File Explorer you can also right-click a PDF → **Open with** → Rough-In Redlines Preview, or double-click a .redline file. It opens in a new tab of the window that's already open.
- Recent files are listed on the Home tab and under **File** → **Open recent**. Right-click one to open its folder or take it off the list.

On the iPad IPAD

- Have the office email you the drawing, then tap it and **Share** → the app. The same works from Files, Teams, OneDrive, AirDrop or any app that can share a PDF.
- Or tap **Open PDF** on the home screen and pick it in Files.
- Photos can be marked up too: **Photos**, or **Take Photo to Mark Up** on an iPad with a camera. Several pictures at once? Choose **Combine into one sheet** or **Separate sheets**.
- Sending a drawing again with the same name asks what to do: **Replace sheet (keep markups)** (when the page count matches), **Replace sheet** or **Keep both**.

On Android ANDROID

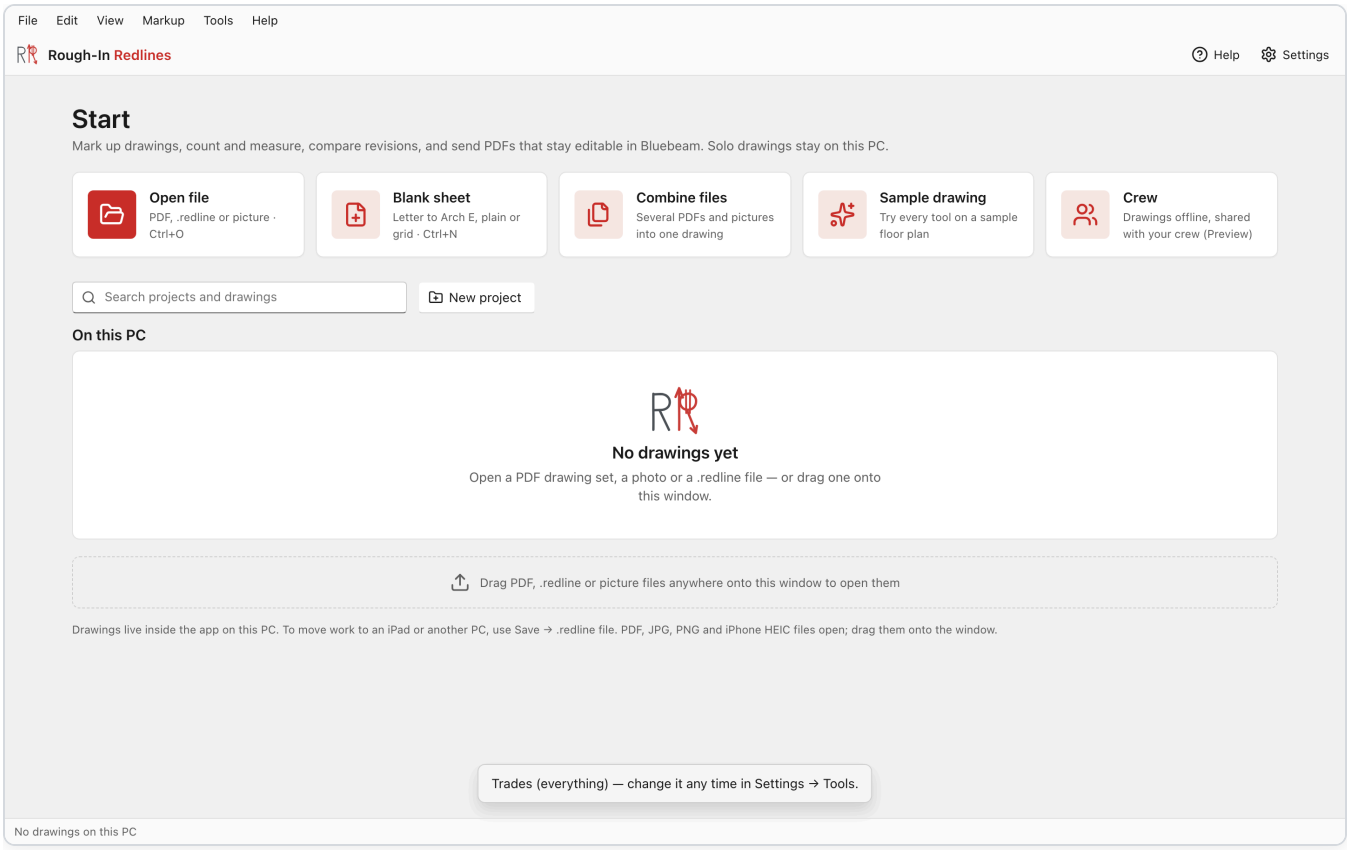
- Tap **Open file** on the home screen and pick the file: Downloads, the tablet, Google Drive, OneDrive or Dropbox.
- From another app (Files, Gmail, Drive, Outlook, Procore): **Open with** → Rough-In Redlines, or **Share** → Rough-In Redlines.
- Recent files show where each one came from. If one has moved: *This file moved or permission was removed — open it again.*

> **PASSWORD-PROTECTED PDFS**
> A PDF that needs a password to open asks for it once; the drawing is then kept in the app without the password. A PDF that only stops printing or copying opens without asking, and those limits don't carry over.

2.6 Try the sample drawing

The quickest way to learn is to play with the built-in sample: a made-up two-sheet electrical floor plan (E1.01 and E1.02, at 1/4" = 1'-0"), with an older revision to try Compare on. It doesn't count as one of your free drawings.

- WINDOWS Click **Sample drawing** on the Home tab (or **File** → **Sample drawing**).
- IPAD Tap **Try a Sample Drawing** on the empty home screen, or **More** → **Add Sample Drawings**.

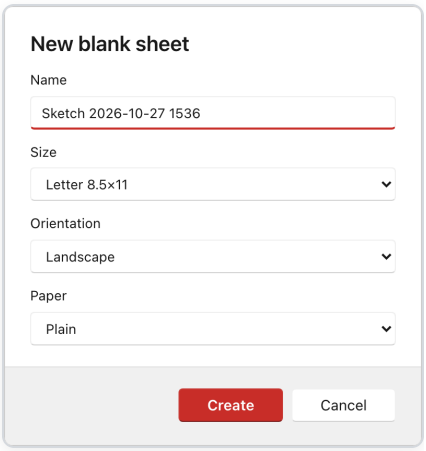


The Home tab the first time the Windows app starts, with the "How will you use…" card.

2.7 Start a blank sheet

For a sketch or a quick detail, start from a blank sheet: **Blank sheet** on the Home tab (`Ctrl` + `N`) on Windows, or **New Blank Sheet** on the iPad.

- **Size**: Letter 8.5×11, Tabloid 11×17, Arch D 24×36 or Arch E 36×48. On Windows also A1, A2, A3 and A4.
- **Orientation**: Landscape or Portrait.
- **Paper**: Plain, Grid ¼" or Grid 1" (the iPad also has Dot Grid).
- It's named **Sketch** with the date and time; rename it if you like.



New blank sheet WINDOWS

3 New in 2.1

Version 2.1 brings Rough-In Crew to everyone, share links from anywhere, one Add button, any color with the APWA utility palette, fills and color legends, borders on pasted pictures, a much better Find & Count and cleaner Bluebeam exports. This chapter walks through each one; the chapters after it describe the rest of the app.

3.1 Where 2.1 runs

Device	Get it from	Rough-In Crew
IPAD iPad and iPhone (and Macs with Apple silicon)	The App Store	Yes. Subscribe in the app ($14.99/month) or use a company license.
WINDOWS Windows 10 and 11	The Microsoft Store, or the installer from roughinredlines.com	Yes. Subscribe in the app ($15/month) or use a company license; owners buy and assign licenses here.
ANDROID Android tablets	Google Play	Not yet. Every drawing tool works; Crew on Android comes later.

3.2 Rough-In Crew: plans and how to start CREW

Crew gives your company shared projects: drawing sets, forms, markups, counts, photos, notes and comments, synced live between everyone's iPhone, iPad and PC (chapter chapter 12). Sign-in is your work email and a 6-digit code we email you; roles are company owner and admin, and on each project project manager, editor or viewer.

What it costs. Crew is **$15 per person per month**, and it **includes Rough-In Pro** (unlimited drawings) on that person's devices while it's active. Each person who works in Crew needs one of these:

- **A personal subscription**: $14.99/month through Apple on iPhone and iPad, or $15/month through Lemon Squeezy on Windows. One subscription covers you on **up to 3 devices** (any mix of PC, iPad and phone), in every company you belong to, whichever store you bought it from.
- **A company license**: the company buys named licenses ($15 per license per month) and assigns one to you. Each license covers **1 to 5 devices**, as the owner sets.

A new company can try Crew free for 7 days with up to 4 people.

Subscribe on iPhone or iPad

1. On Home tap **Crew** (the people icon), enter your work email, tap **Email me a code** and type the code.
2. Where a project is read-only, tap **Subscribe…** on the banner, then **Subscribe — $14.99/month**, and confirm with your Apple Account.
3. The banner goes away and you can mark up. Settings → **Subscription** shows "Crew subscription · renews …". **Restore Purchases** brings a subscription back on a new iPad or iPhone; **Manage**

subscription opens Apple's page to change or cancel it.

Subscribe on Windows

1. Open **Crew** and sign in with your work email and the emailed code.
2. Click **Subscribe — $15/month** (on the no-license card or the read-only banner). The secure Lemon Squeezy checkout opens in your browser; pay there.
3. Back in the app click **I've paid — check again** (it also checks when you come back to the window). The account menu → **Crew subscription…** shows the status and your devices; **Manage billing** changes the card or cancels.

Buy and assign company licenses (owner, on Windows)

1. Sign in to Crew on a Windows PC as the company's owner. Click **Start Crew — $15 per person per month**, enter how many licenses you need and pay on the checkout page.
2. Company ⋯ menu → **Licenses…**: assign each license to a person by email. Set **Devices per license** (1–5). Lowering it signs out each person's least recently used devices over the new limit.
3. **Buy licenses** adds more; **Manage billing** changes the card, shows invoices or cancels. On iPad and iPhone your people just sign in; licenses are shown there but assigned on Windows.

GOOD TO KNOW

- **A lapsed plan is view-only, not deleted.** If your subscription ends and no license covers you (or the company's plan lapses), you can still open and view Crew projects but not change them. Changes you make on the device meanwhile wait there and go through when you're covered again.
- **Too many devices?** Adding a fourth device to a subscription (or more than a license allows) asks you to remove one first.
- **Cancel** through Apple (Manage subscription, or Settings → your name → Subscriptions on the device) or Lemon Squeezy (Manage billing). You keep Crew to the end of the month you paid for.
- **Deleting your Crew account does not cancel billing.** Settings → **Crew account** → **Delete account…** reminds subscribers to cancel first; do that, or the store keeps charging.
- A license and your own subscription are two separate charges. If your company gives you a license, cancel the subscription you no longer need.

3.3 Share links from Home and Crew, as one PDF, at the size you need

CREW

Share links (chapter 12) now start from more places:

- **Share files** on Home: a link of drawings and files on this device, no Crew project needed. Also on the Crew home screen.
- On a drawing or sheet: **Share link…** (the file as it is) or **Share link with markups…** (a flattened PDF with your markups). On a set: **Share link to this set…**.

1. Pick what goes in: **Add** (Photos, Take photo, Files) or **Add sheets from this project…**.

2. **Send as one PDF** (on by default): a whole set goes as one PDF. When the set has no markups, it is the original file itself. Turn it off for one file per sheet.
3. **File size**: **Actual size**, **Smaller (good for email)** or **Smallest**. The estimate shows first ("38 MB → about 6 MB"). Only photos and scanned pictures get smaller; linework, text and markups stay sharp, and the drawing on your device isn't changed. Your choice is remembered.
4. **Link works for** 1, 7 or 30 days or **Until…** a date; **Downloads**: Unlimited, One-time or Limited; **PIN (optional, 4–6 digits)**, sent separately.
5. **Create link**, then copy it. The link is shown only this once.

Shared links (Crew home) lists the links with their downloads: **Extend…** moves the end date (a year at most); **Revoke link** stops it at once. The same **File size** choice is offered on Send PDF (flattened and Bluebeam / Acrobat, when the drawing has pictures worth shrinking) and Send to office.

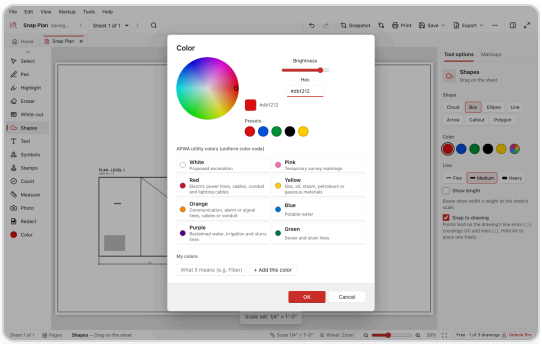
3.4 One Add button

Everywhere you bring something in (Home, a project on this device, a Crew project's **Add drawings**, and Share files) there is one **Add** button with the same three choices, in this order:

- **Photos**: pick one or several from your photo library. When adding drawings, **Photo size** lets you make them smaller first.
- **Take photo**: the camera (on devices that have one).
- **Files**: PDFs and pictures from Files, OneDrive, Dropbox and so on.

3.5 Any color and the APWA utility colors

1. Select a markup or pick a tool. In the color row, choose the rainbow swatch (**More colors…**).
2. Pick from the color wheel, type a **Hex** value (for example *#ff8200*), or tap a **Recent** color.
3. Or pick an **APWA utility color**: each shows what it means (table below).
4. To keep a color, type what it means (for example *Fiber*) and **Add this color** to **My colors** (up to 24, on this device).
5. Click **OK** (IPAD **Use**).



More colors… WINDOWS

Any markup can take any color: text, symbols, counts, shapes, stamps and measurements. Older versions of the app show the nearest of the five standard colors.

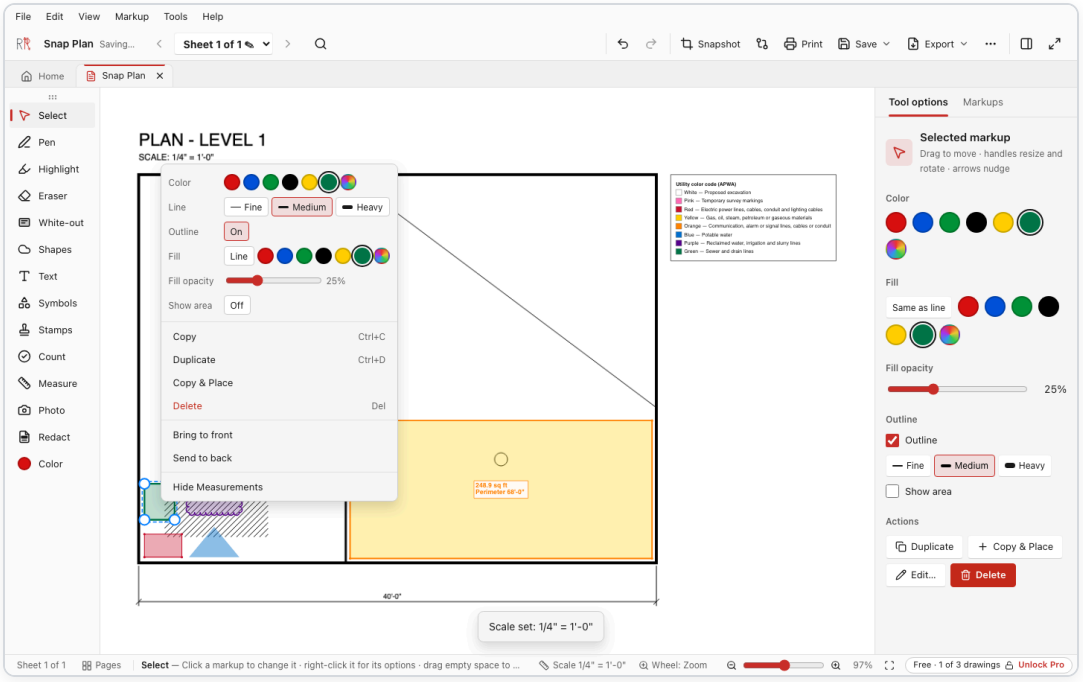
APWA color	Means
White	Proposed excavation
Pink	Temporary survey markings
Red	Electric power lines, cables, conduit and lighting cables

APWA color	Means
Yellow	Gas, oil, steam, petroleum or gaseous materials
Orange	Communication, alarm or signal lines, cables or conduit
Blue	Potable water
Purple	Reclaimed water, irrigation and slurry lines
Green	Sewer and drain lines

3.6 Fills, polygons and the area outline

1. For a plain shaded shape, pick **Shapes** → **Polygon** and draw it like an area: click (tap) each corner, then click the first corner again, double-click or press `Enter`. A polygon needs no scale.
2. Select a polygon, area, box, ellipse or cloud. Under **Fill** pick a color (**Own** = the markup's color, or **No fill** on boxes, ellipses and clouds) and slide **Opacity** (up to 80 %).
3. Turn **Outline** on or off and set its **Weight**.
4. On an area measurement, **Show area** turns the area and perimeter label on or off. Off makes it a plain shaded polygon.

WINDOWS The same settings are in the side panel and on the right-click menu. IPAD Select the shape, then **Fill** in its bar.



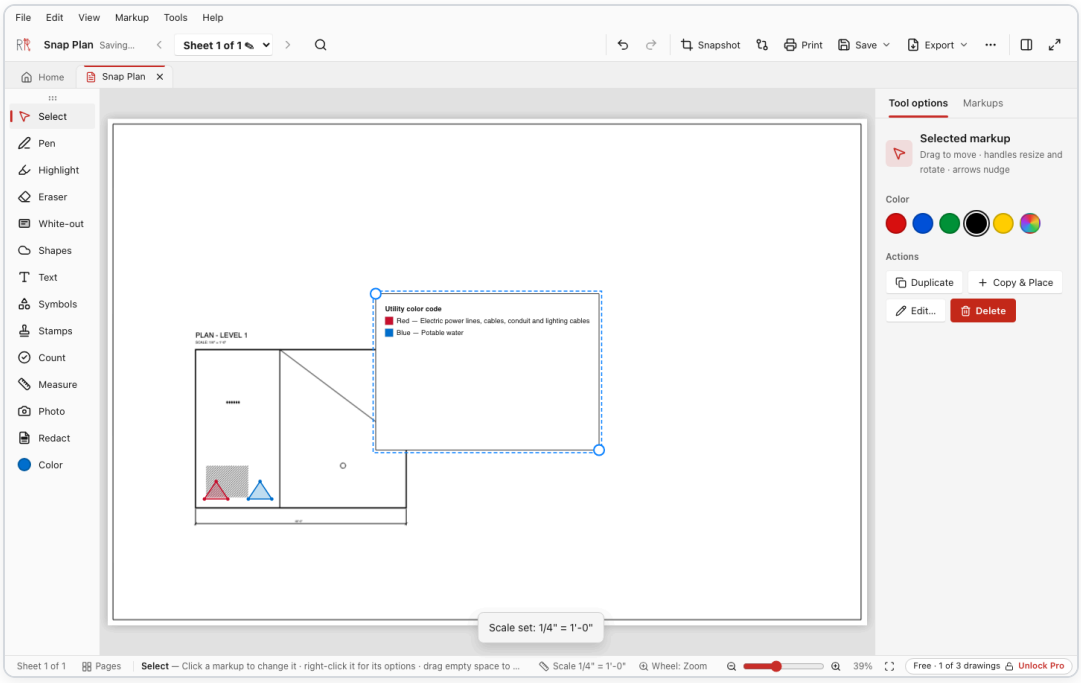
Fill and outline options for a polygon WINDOWS

3.7 Color legends and the Color key

1. **Tools** → **Insert color legend…** WINDOWS, or More (⋯) → **Insert color legend…** IPAD.

2. Give it a **Title** (for example *Utility color code*). Choose the APWA colors or your own (My colors), and edit each label.
3. Tick **Only colors used on this sheet** to list just the colors that appear on that sheet (it updates as you mark up).
4. Insert it, then drag it into place and drag its corners to size it. Select it → **Edit legend…** to change it later.

Legends print and export (Bluebeam sees a box with the legend's text). **Show color legends** (same menu) hides or shows them all. **Color key** opens an on-screen list of the palette's colors and meanings; it never prints.



A legend listing only the colors used on the sheet **WINDOWS**

3.8 Pasted pictures: border and stacking

A snapshot you paste on the sheet (5.10) can have a border, so a blown-up detail stands out:

1. Select the pasted picture and turn **Border** on.
2. Pick **Thin**, **Medium** or **Heavy**. The color swatch now sets the border's color (any color, including APWA and My colors).
3. To border every new snapshot automatically, turn on **Border new snapshots** in Settings (next to the snapshot preview setting).

Arrows, clouds, notes, symbols and pen marks drawn on a pasted picture stay on top of it, on screen, in print and in Bluebeam. **Send to back** and **Bring to front** change the order. The eraser, Symbols and Stamps leave the picture alone: select it and press Delete to remove it.

3.9 Find & Count, improved

- **Reads tag text**: tags with different letters or numbers (also letters drawn as outlines) are told apart.

- **Any angle**, and symbols with a pipe, centerline or dashed gridline running through them are still found.
- A **progress bar** while it searches; matches join the list as they're found, and **Cancel** keeps what was found so far.
- In a Crew set view, the box is used on the sheet you're looking at.

How to use it: chapter 7.

3.10 Bluebeam exports keep counts, highlights and fills

In **PDF for Bluebeam / Acrobat**:

- **Counts** arrive as Bluebeam **Count Measurements**, grouped with their totals, from the iPad as well as Windows, so the PM can count, move and edit them in Revu. Edited marks stay check marks.
- **Highlights** stay see-through highlights when moved or recolored in Bluebeam.
- **Polygons** arrive as Bluebeam polygons with their fill color and transparency; filled boxes, ellipses and clouds keep their fill; areas with a scale stay Area Measurements, now with their fill.
- A pasted picture's border comes along as a box around the picture.

4 Finding your way around

The home screen lists your drawings; the editor is where you mark them up. This chapter shows the parts of each screen and the settings that change how the app looks and works.

4.1 The Home tab WINDOWS

The Home tab is the start page. Get back to it any time with the **Home** tab, **File** → **Home** or `Ctrl` + `Shift` + `H`.

- **The tiles**: **Open file**, **Blank sheet**, **Combine files** (several PDFs and pictures into one drawing), **Sample drawing**, **Crew** CREW.
- **Recent**: the files you opened lately (name, folder, date opened, sheets). Right-click one for **Open**, **Open file location** or **Remove from list**; **Clear list** empties it.
- **On this PC**: every drawing kept in the app, as cards with a thumbnail, the name and how many sheets. Click a card to open it. The trash button or right-click → **Remove from this PC…** deletes it and its markups (save a .redline copy first if you might need it).
- **Projects on this device** V2: group drawings by job. See 10.7.

Drawings live inside the app on this PC. To move work to an iPad or another PC, use **Save** → **.redline file** (see 11.8).

4.2 The editor WINDOWS



The Windows editor with the sample drawing: menu bar and command bar at the top, tabs, the toolbar on the left, the side panel on the right and the status bar along the bottom.

Part	What it does
Menu bar	**File**, **Edit**, **View**, **Markup**, **Tools** and **Help**. Every command is here with its shortcut. Press `Alt` or `F10` to use it from the keyboard (`Alt` + `F` opens File).
Command bar	The drawing's name and whether it is **Saved**; ‹ **Sheet 1 of 2** › to change sheets (click the name for the sheet list); Find (magnifier); Undo and Redo; **Fill form** (only on a PDF with form fields); **Snapshot**; Compare; **Print**; the **Save**, **Export** and ⋯ (More) menus; the side panel button and **Focus**.
Tabs	Every drawing you open gets a tab. Click one, or press `Ctrl` + `Tab`, to switch; each comes back on the same sheet and zoom with its own undo history. `Ctrl` + `W`, the × or a middle-click closes it. A dot on a tab means it has changes since it was last saved as a .redline file.
Toolbar	The markup tools: Select, Pen, Highlight, Eraser, White-out, Shapes, Text, Symbols, Stamps, Count, Measure, Photo, Redact, and **Color**. Tools that don't fit go under **…**.
Side panel	**Tool options** for the tool you picked (color, width, shape kind and so on), **Markups** (every markup on the sheet), and **Search** while Find is open. Hide it with `\`; drag its inner edge to make it wider.
Status bar	The sheet you're on (click to go to another), the **Pages** button, what the tool does, the sheet's scale (**No scale set** until you set one), the mouse wheel mode, and the zoom slider, percentage and Fit button.

Scrolling through a set

A set opens in **Continuous scroll**, like Acrobat or Bluebeam: the sheets are one under the other. Turn the mouse wheel to scroll; `Ctrl`+wheel (or a touchpad pinch) zooms at the pointer. `Page Down` / `Page Up` move a window at a time; `Home` and `End` go to the top and bottom of the set. Every tool works on the sheet under the pointer. To pan, drag with the middle or right button, or hold `Space` and drag.

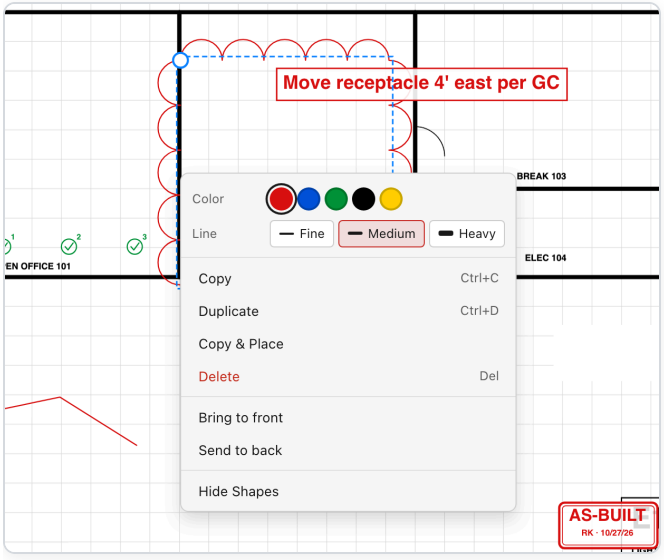
Prefer one sheet at a time, like the iPad? **View** → **Page display** → **One page at a time** (or the page button in the status bar). Then the wheel zooms and Page Down / Page Up change sheets. To always have the wheel scroll or always zoom, use **View** → **Wheel: Scroll pages** or **Wheel: Zoom** (`Ctrl`+wheel then does the other one).

More room for the drawing

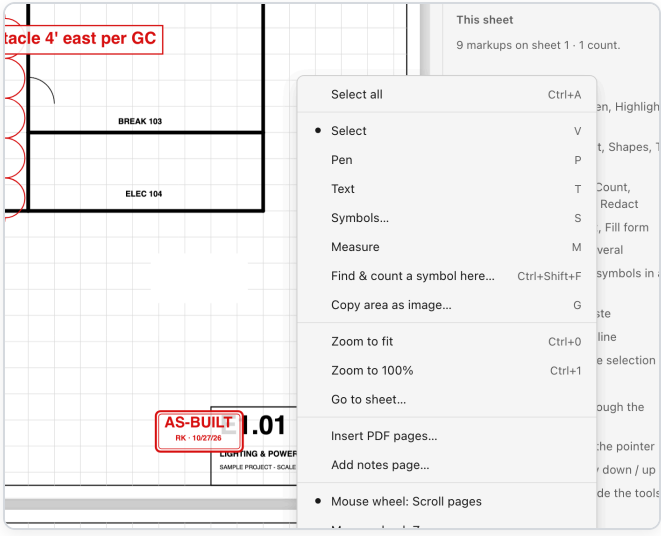
- **Focus** (`F`, or the Focus button at the top right) hides the tools; the pen still draws. Press `F` again or click **Show tools**.
- **Full screen**: `F11`.
- **Toolbar position**: drag the toolbar by its grip (the dots at its start) to any edge, or **View** → **Toolbar position** (Bottom, Top, Left or Right). **View** → **Toolbar icons** picks Small or Large buttons, and **Show labels** turns the names on or off.

Right-click for options

Right-click any markup (with any tool) for its options right where it is: color, line weight, text size and so on change as you click, and below them **Edit text…**, **Copy**, **Duplicate**, **Copy & Place**, **Delete**, **Bring to front** / **Send to back** and more. Right-click empty sheet for **Paste**, **Select all**, the tools, zoom and **Go to sheet**. `Shift`+`F10` or the Menu key opens it for the selection.



Right-click a markup.

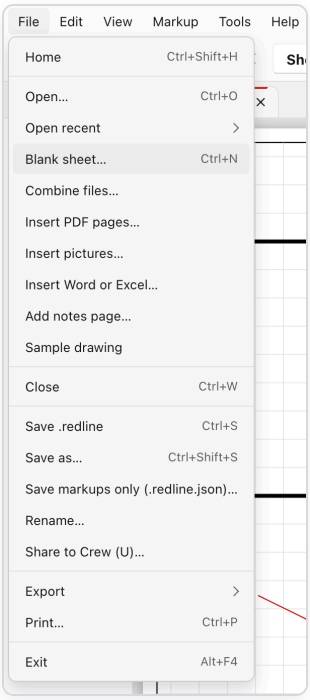


Right-click empty sheet.

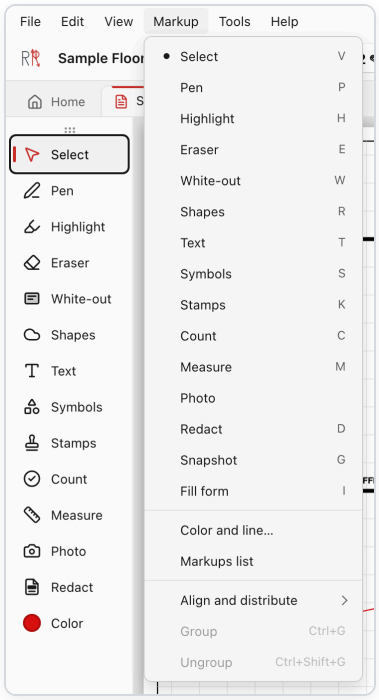
4.3 The menus WINDOWS

Items for tools you've turned off in Settings → Tools are hidden.

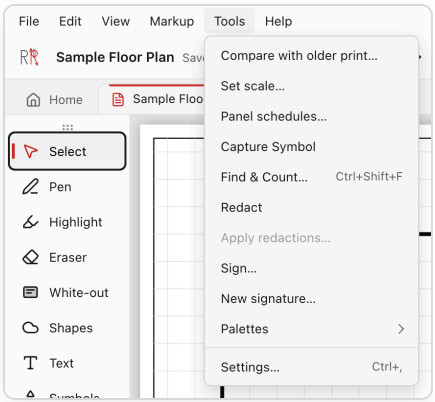
Menu	What's in it
File	Home · Open… · Open recent · Blank sheet… · Combine files… · Insert PDF pages… · Insert pictures… · Insert Word or Excel… · Add notes page… V2 · Sample drawing · Close · Save .redline · Save as… · Save markups only (.redline.json)… · Rename… · Share to Crew… CREW · Export ▸ · Print… · Exit
Export ▸	Bluebeam / Acrobat PDF… · Flattened PDF… · Marked-up sheets only… · Password-protected PDF… · Save a smaller copy… · Filled form, flattened… · Counts CSV… · Picture as JPG…
Edit	Undo · Redo · Cut · Copy · Paste · Duplicate · Delete · Select all · Find… · Find next · Find previous · Copy & Place · Edit markup… · Bring to front · Send to back
View	Zoom in · Zoom out · Fit to window · Actual size (100%) · Next sheet · Previous sheet · Go to sheet… · Pages… · Page display ▸ · Wheel: Automatic / Scroll pages / Zoom · Next tab · Previous tab · Side panel · Focus (hide the tools) · Full screen · Toolbar position ▸ · Toolbar icons ▸ · Show labels · Theme ▸ (Light, Dark, Match Windows)
Markup	Every tool with its key · Color and line… · Markups list · Align and distribute ▸ · Group · Ungroup
Tools	Compare with older print… · Set scale… · Panel schedules… · Capture Symbol · Find & Count… · Redact · Apply redactions… · Sign… · New signature… · Palettes ▸ · Settings…
Help	How to use (`F1`) · Keyboard shortcuts (`Ctrl`+`/`) · Check for updates… · About



File



Markup



Tools

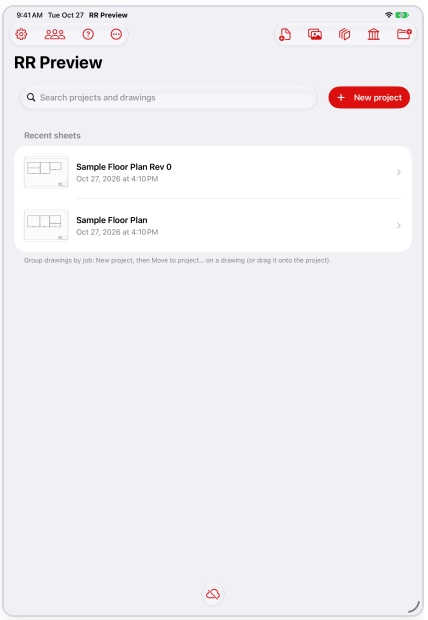
How to use (`F1`) opens the built-in help topics, with a **Search help** box. They are a quick reminder of everything in this manual.

4.4 The iPad home screen IPAD

The buttons along the top: on the left **Settings**, **Crew** CREW, **How to use** and **More**; on the right **New Blank Sheet**, **Take Photo to Mark Up**, **Photos**, **Combine files**, Procore and **Open PDF**. On an iPhone most of them are in a **More** menu next to Open PDF.

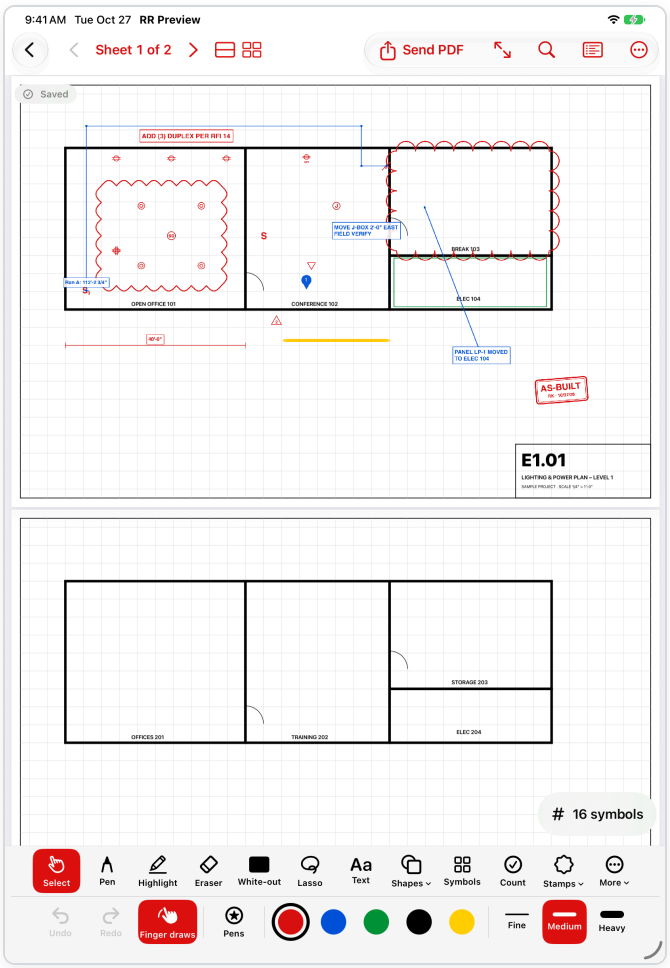
Below are your drawings (**Recent sheets**), and in Version 2 your **Projects on this device** with **Search projects and drawings** and **New project** (see 10.7). Swipe a drawing to **Delete** it; long-press it for **Compare with Older Print…** and **Share to Crew….**

Settings → **Backup** keeps a copy of every drawing in a cloud folder; see 11.9.



The iPad home screen with the sample drawing and its older revision.

4.5 The iPad editor IPAD



The iPad editor with the toolbar at the bottom.

Top bar, left to right: Back to the home screen; ‹ **Sheet n of N** › (tap the label for the **Sheets** list with thumbnails); the page display button (**Continuous scroll** / **One page at a time**) and **Pages**; the Crew menu on a Crew sheet; **Send PDF**; **Full Screen**; **Find**; **Markups**; and **More (···)**.

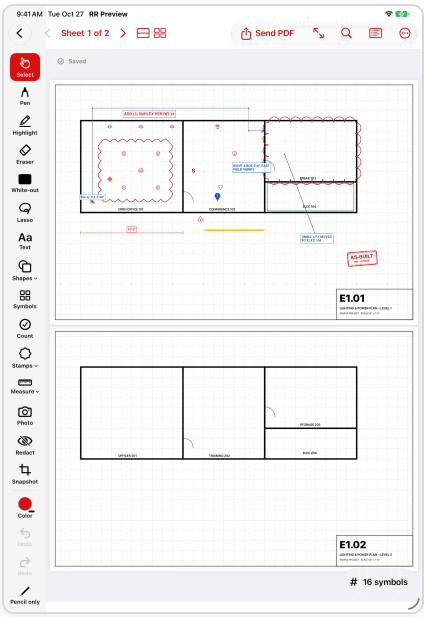
More (···) holds: Save Now · Capture Symbol · Find & Count… · Snapshot (copy area as image) · Paste · Set Scale… · Redact · Pages… · Insert PDF pages… · Insert pictures… · Add notes page… · Compare with Older Print… · Panel Schedules… · Send to office… / Share to Crew… · Settings · How to use.

The toolbar has two rows. The tools: **Select** · (**Fill** on a PDF with form fields) · **Pen** · **Highlight** · **Eraser** · **White-out** · **Lasso** · **Text** · **Shapes** · **Symbols** · **Count** · **Stamps** · **Measure** · **Photo** · **Redact** · **Snapshot**. Below them: **Undo**, **Redo**, the finger button (**Finger draws** / **Pencil only**), your saved pens with **Save Pen**, the colors (red, blue, green, black, yellow), the widths **Fine**, **Medium** and **Heavy**, and the opacity slider for the highlighter. Tools that don't fit go under **More**. The buttons are big enough to use with gloves.

Put the toolbar where it suits you: Settings → Appearance → **Toolbar position** (Bottom, Top, Left or Right), or long-press an empty part of the toolbar → **Move toolbar**. On the left or right the tools stack in one column, and **Color** opens a pop-out with the colors, line widths and saved pens.

A **three-finger tap** switches Full Screen on and off. The counter at the bottom (**16 symbols**) opens the counts.

On an iPhone the toolbar has one row: Select · Pen · Highlight · Text · Symbols · More, and a second row with Undo, Redo, finger, color, width and saved pens. **More** opens big tiles for everything else. **Send PDF** is just **Send**, and Left / Right toolbar positions use the bottom.



The toolbar on the left.

4.6 On an Android tablet ANDROID

The Android app looks like the Windows app without the menu bar.

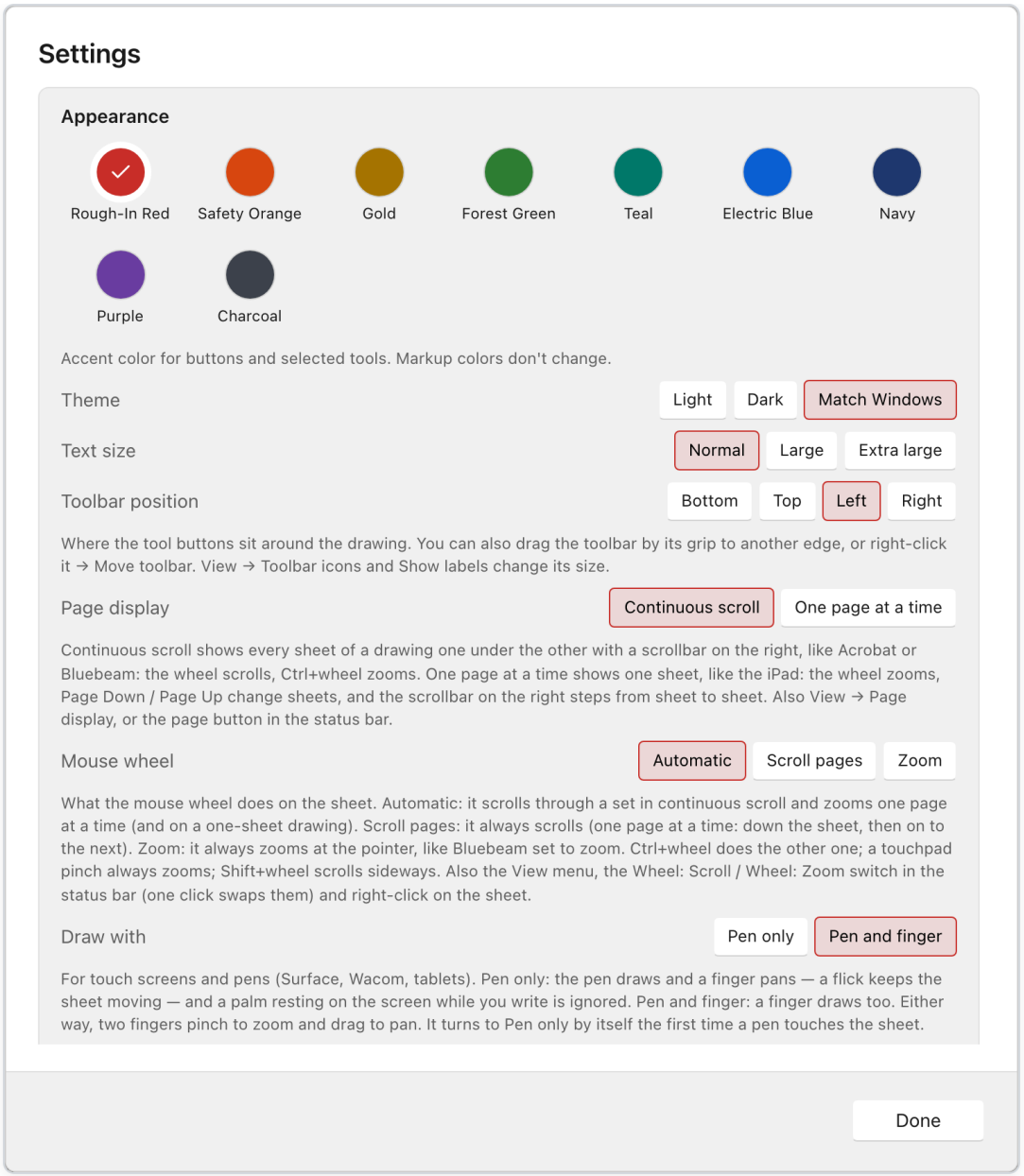
- The **command bar** at the top has Open, Undo / Redo, Save, Export, Print, Compare, Snapshot, Find, Focus and the side panel button. Its ⋯ button at the top right holds what is in the Windows menus: Pages…, Find & Count…, Panel schedules… and so on.
- **Long-press** a markup for its options (what a right-click does on Windows); long-press empty sheet for Paste, Select all, Insert PDF pages…, the tools, zoom, Go to sheet, Copy area as image… and Find & count a symbol here….
- To select several markups, hold still on empty paper for a moment and drag a box, or choose **Select several** (the first item of a long-press menu): taps then add or remove markups, and a bar offers **Align** ▸, **Group** / **Ungroup**, **Delete** and **Done**.
- The **Back** button closes the open menu or window, then cancels what you're doing, then goes from a drawing to the home screen. On the home screen it puts the app in the background; your work is kept.
- In portrait the command bar shows icons only. Theme offers **Match system** instead of Match Windows.

4.7 Settings

WINDOWS **Tools** → **Settings…** (`Ctrl`+`,`) or the Settings button on the Home tab. IPAD the **Settings** button on the home screen, or More → Settings.

Section	What you can set
Rough-In Pro	Status, how many drawings, unlock, license key, Check now, Deactivate this PC (2.4).

Section	What you can set
Appearance	Accent color (Rough-In Red, Safety Orange, Gold, Forest Green, Teal, Electric Blue, Navy, Purple, Charcoal; markup colors don't change). Theme (Light, Dark, Match Windows). Text size (Normal, Large, Extra large). Toolbar position. Page display. Mouse wheel (Automatic, Scroll pages, Zoom). **Draw with**: Pen only or Pen and finger. **Show preview after Snapshot**. **Reopen drawings from last time**. WINDOWS
Tools	Which tools and features show (4.8).
Your initials	Up to 4 letters. Added to stamps and to every markup you make, so the office can see who marked what.
Company profile	Company name, logo and office addresses, printed on panel schedules. Kept on this device.
Symbol libraries	Which libraries show in the symbol panel (6.2).
Links	**Link sheet references**: sheet numbers written on a drawing (3/E5.01, SEE E6.01) become links you can click with Select.
Export	**Include symbol count page** at the end of flattened PDFs. IPAD also **Include panel schedules**.
Procore CREW	**Procore** on or off. Off hides Add from Procore, Connect Procore and Send to Procore.
Pens IPAD	**Draw with finger** and **Restore default saved pens**.
Measure IPAD	**Snap measurements** and **Show preview after Snapshot**.
Backup IPAD	Back up every drawing to a folder in OneDrive, SharePoint, Google Drive, iCloud Drive or Dropbox (11.9).
About	Version and edition, How to use, website, support and privacy links.



Settings on Windows: Appearance.

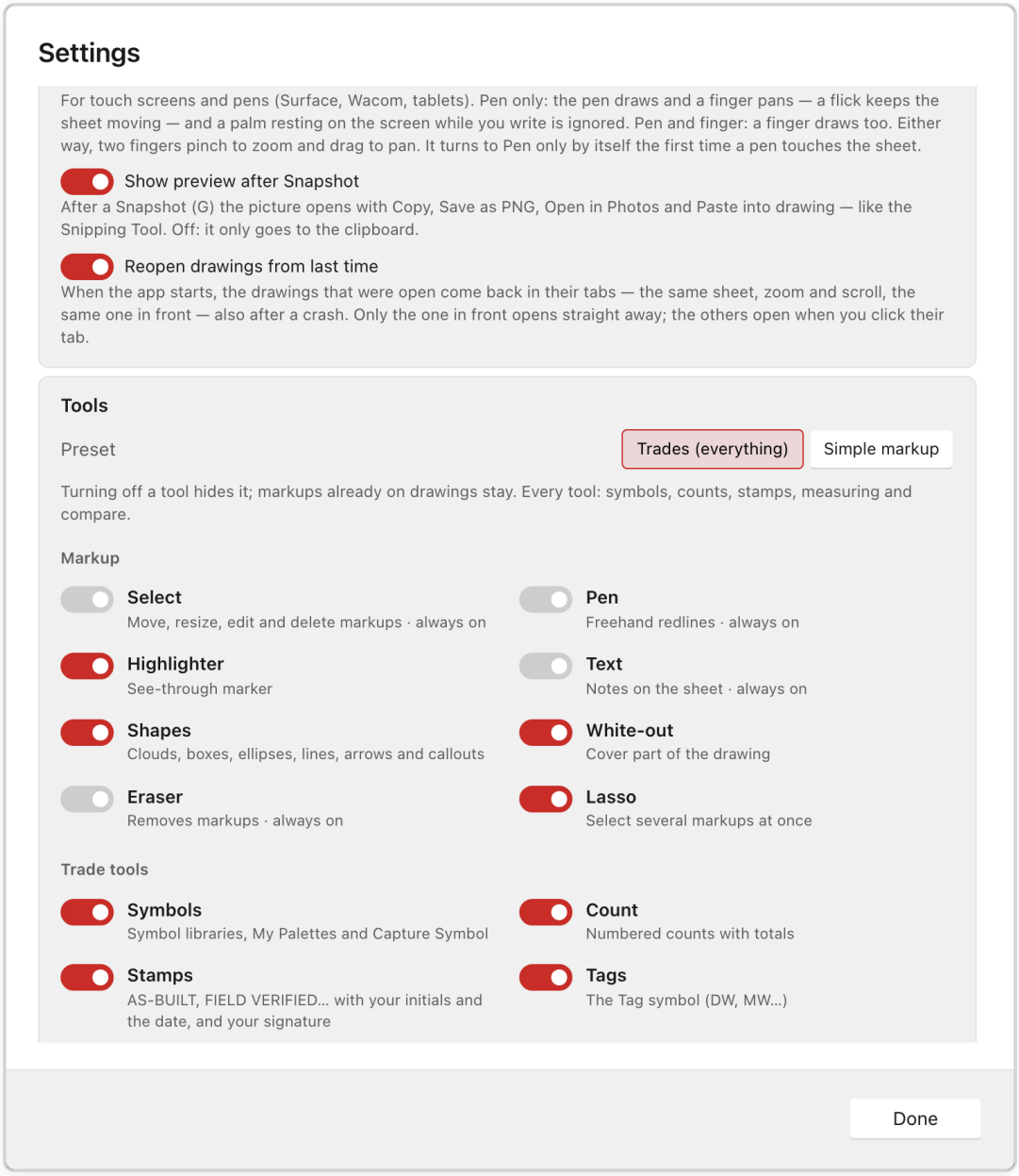
4.8 Turn tools on or off

Settings → **Tools** chooses which tools you see, so the toolbar shows only what you use. Pick a preset at the top:

- **Trades (everything)**: the default. Every tool.
- **Simple markup**: Pen, Highlighter, Text, Shapes, White-out, Eraser, Lasso, Photos and the flattened and Bluebeam PDFs. No Symbols, Count, Stamps, Tags, Measure or Compare. Good for office staff who only review.
- **Custom** appears by itself when you flip switches to make your own mix.

The switches are grouped as **Markup**, **Trade tools**, **Measuring**, **Drawings** (Compare revisions, Photos on the sheet, Blank sheet / pictures as drawings, Pages and combining PDFs, Redact) and **Sending** (Flattened PDF and Print, Bluebeam / Acrobat editable PDF, Redline Copy (.redline), Counts CSV). Select, Pen, Text, Eraser and Flattened PDF / Print are always on.

> **NOTHING IS LOST**
> Turning off a tool only hides it: its button, menu items, panel section and shortcut. Markups already on drawings stay; they still show, print and export, and you can still select, move or delete them.



Settings → Tools and Symbol libraries.

5 Marking up

Pick a tool, pick a color, and draw. Every markup carries your initials and the date, and is saved as you go. A sheet always opens with Select, so the first click never draws by accident.

5.1 Select, move and change markups

Select (`V`) clicks a markup to move, resize, rotate, edit or delete it. Drag the round handles to resize; drag the orange dot to turn a symbol, stamp or signature. The arrow keys nudge the selection a little, `Shift`+arrow ten times as far. Double-click a note (or any markup with text) to edit it.

- **Several at once**: WINDOWS `Shift`+click, or `Shift`-drag a box around them (the Lasso). `Ctrl`+`A` selects every markup on the sheet. IPAD the **Lasso** tool circles ink and markups together. ANDROID hold still on empty paper and drag a box, or **Select several**.
- With something selected, the side panel shows its **Color** and the **Actions**: **Duplicate**, **Copy & Place**, **Edit…**, **Group**, **Delete**.
- A box, ellipse or cloud is picked (and erased) by its outline, so a click inside one gets the symbol, note or count mark in it. Once a box is selected, drag it from anywhere inside.
- IPAD Long-press a markup for **Delete**, **Copy**, **Duplicate**, **Copy & Place**, **Bring to Front**, **Send to Back** and more. A selected symbol can be pinched to resize and twisted to rotate (it snaps to right angles).

Copy, paste, duplicate and Copy & Place

`Ctrl`+`X` / `Ctrl`+`C` / `Ctrl`+`V` cut, copy and paste markups, also onto another sheet or drawing. **Duplicate** (`Ctrl`+`D`) makes a copy next to it. **Copy & Place** takes a sized and turned markup and drops a copy at every click until you press **Done (Esc)**: handy for the same fixture in every room.

Undo and redo

WINDOWS `Ctrl`+`Z` undoes, `Ctrl`+`Y` (or `Ctrl`+`Shift`+`Z`) redoes, or use the arrows on the command bar. Each tab keeps its own history. IPAD the **Undo** and **Redo** buttons on the toolbar (`⌘`+`Z` with a keyboard). Undo also covers page changes, counts made by Find & Count, and redacting a drawing until you close it.

5.2 Pen and highlighter

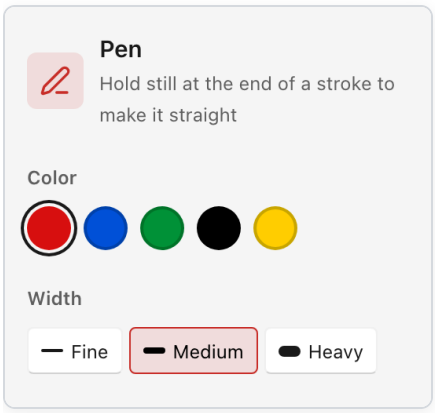
Pen (`P`) and **Highlight** (`H`) draw freehand. Pick a color (red, blue, green, black or yellow) and a width: **Fine**, **Medium** or **Heavy**. For the highlighter, the **Opacity** slider sets how see-through it is (100% is solid).

> STRAIGHT LINES
> Hold the pen still for half a second at the end of a stroke and it snaps straight.

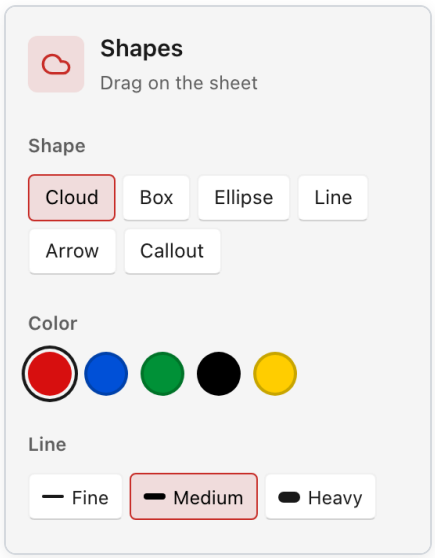
- WINDOWS A Surface or Wacom pen draws thicker the harder you press, and its eraser end erases with any tool. On a touch screen, Settings → Appearance → **Draw with** chooses **Pen only** (a finger pans, a

resting palm is ignored) or **Pen and finger**. It switches to Pen only by itself the first time a pen touches the sheet.

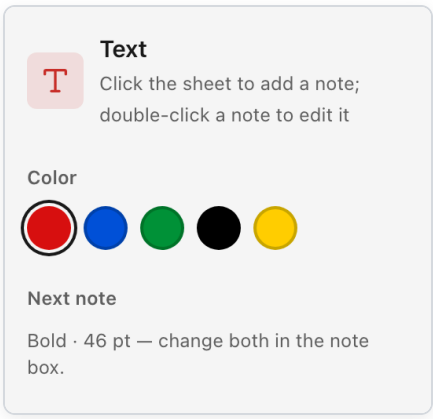
- IPAD With Apple Pencil, the Pencil draws and one finger pans; pinch to zoom. Without a Pencil, turn on **Finger draws**: one finger draws and two fingers pan. It turns off by itself the first time a Pencil is used.
- IPAD **Saved pens** keep a tool, color, width and opacity together ("Red Fine", "Yellow 50%"). **Save Pen** adds the current one (up to 8); long-press a saved pen to remove it. Settings → Pens → **Restore default saved pens** brings back the starting set.
- ANDROID Same as Windows: Settings → Appearance → **Draw with**. The S Pen's side button: hold it and tap a markup for its options, or hold it and drag to pan.



Pen options WINDOWS



Shapes options



Text options

5.3 Eraser

Eraser (`E`) removes whatever it touches: pen and highlighter strokes, shapes, white-outs, symbols, count marks, text, stamps and measurements. One Undo puts it back. Pick its **Size**: Small, Medium or Large. Photo pins are kept; select one and press `Delete`. The eraser only removes *your* markups; to hide part of the drawing itself, use White-out.

5.4 Shapes: clouds, boxes, lines, arrows and callouts

Pick **Shapes** (`R`), choose the kind, then drag on the sheet.

Windows	iPad	Use it for
Cloud	**Revision Cloud**	Around a change. One change, one cloud.
Box	**Rectangle**	An area, a panel, a room.
Ellipse	**Ellipse**	Circling an item.
Line	**Line**	A straight run or a leader.
Arrow	**Arrow**	Pointing at something.
Callout	**Arrow Callout**	A note with an arrow: drag from the point to where the text goes.

Pick the color and **Line** weight (Fine, Medium, Heavy). Drag the round handles to resize, or to move a callout's tip.

Show length V2: lines and arrows show their length at the sheet's scale, boxes their width × height, circles their diameter. It needs a scale (see chapter 8); until then the panel says *Set a scale to see lengths*. For lines, arrows and boxes, **Snap to drawing** makes the ends land on the drawing's linework.

5.5 Text notes

Pick **Text** (`T`), click where the note goes, and type. In the note box:

- **Font**: Bold, Regular, Serif, Mono, Marker or Handwriting.
- **Size** in points on the printed sheet: S 8, M 12, L 18, XL 28, XXL 48, or type any size from 1 to 200. **Drawing** matches the size of the drawing's own text: handy on a big drawing reduced onto Letter.
- **White box behind the text** keeps a note readable over linework.
- WINDOWS `Ctrl`+`Enter` or the button saves; `Enter` adds a line.

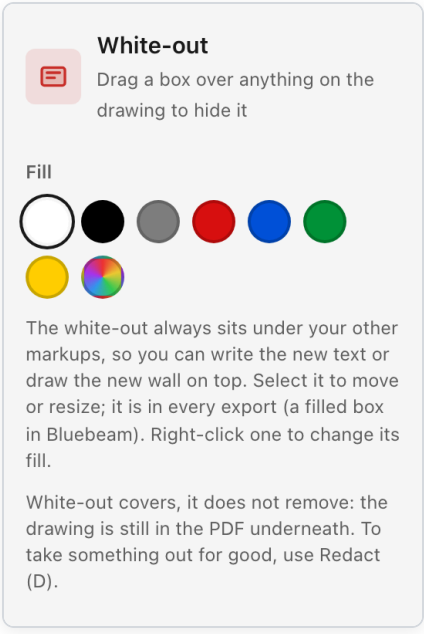
Double-click a note to edit it. New notes start at the last size you used.

5.6 White-out

White-out (`W`) hides part of the drawing (an old wall, a note that no longer applies) under a box: drag it over what should go, then write or draw the new work on top. Pick another fill in the panel, or right-click a white-out → **Fill**: White, Black, Gray, a markup color or **More colors…**.

A white-out covers everything under it: the drawing and the markups drawn before it. What you draw after it stays on top. Right-click it → **Send to back** to put it under your markups again.

> **WHITE-OUT COVERS, IT DOESN'T REMOVE**
> The drawing is still in the PDF underneath, and someone with Acrobat or Bluebeam can select and search it. To take something out for good, use **Redact** (9.2).



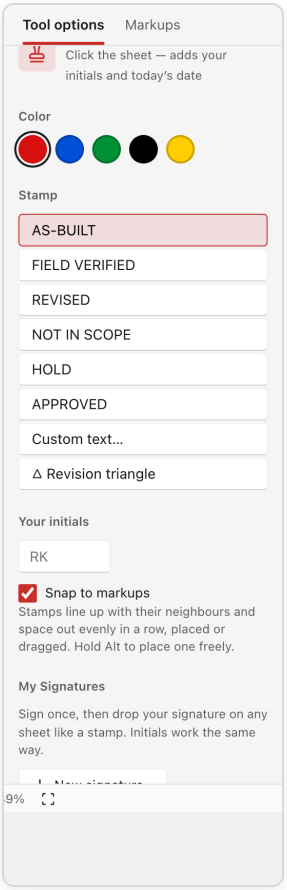
White-out options

5.7 Stamps and the revision triangle

Pick **Stamps** (`K`), choose a stamp and click the sheet. Each stamp adds your initials and today's date.

- The stamps: **AS-BUILT**, **FIELD VERIFIED**, **REVISED**, **NOT IN SCOPE**, **HOLD**, **APPROVED** and **Custom text…** (your own words).
- △ **Revision triangle**: a triangle with a number, placed with one click. Each one on a sheet takes the next number (or the next letter once you've used letters). Double-click one to change it. Group it with its cloud (`Ctrl` + `G`) so they move together.
- **Your initials** are set in the panel or in Settings.
- **Snap to markups** lines stamps up with their neighbors (5.11).

Below the stamps are **My Signatures** (next section).

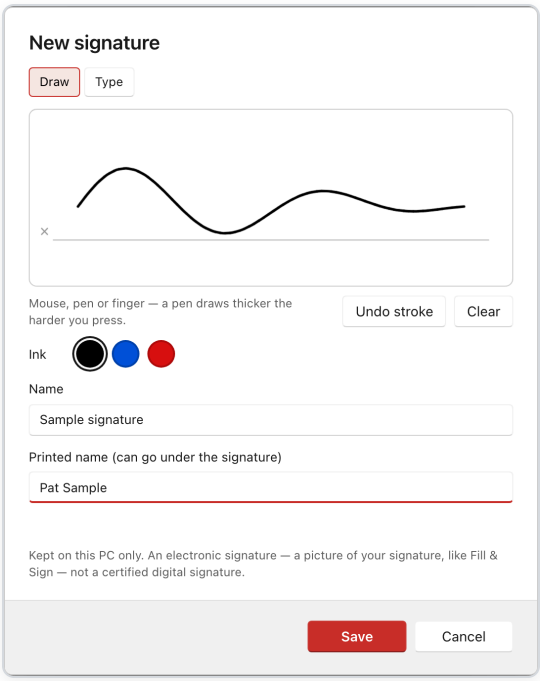


Stamps WINDOWS

5.8 Signatures

Sign once, then drop your signature on any sheet like a stamp.

1. Stamps tool → **My Signatures** → **New signature…** (or **Tools** → **New signature…**).
2. Sign in the box with the mouse, a pen (it draws thicker the harder you press) or your finger. **Undo stroke** and **Clear** start over. Or choose **Type** to type your name in one of three script styles.
3. Pick the **Ink** (black, blue or your markup color), give it a **Name** ("Randy – signature", "RK – initials") and, if you want it under the signature, the **Printed name**. Click **Save**.
4. To sign: pick it in the Stamps panel (or **Tools** → **Sign…**) and click the sheet. It goes down about 2½" wide, with a date line and your printed name when those are ticked.



The New signature window.

Move, size and turn it like any stamp. To pin it in place, right-click it → **Lock**: a locked signature can't be dragged, sized or erased by accident. Right-click for **Unlock**, **Add date** / **Remove date**, **Printed name…**, **Smaller** / **Bigger** and **Delete**. IPAD long-press it for the same, plus **Rotate 90°**.

Signatures are never counted. Your signatures stay on this device only; they leave it only on a drawing you export, print or send.

> **ELECTRONIC, NOT DIGITAL**
> This is an electronic signature (a picture of your signature, like Acrobat's Fill & Sign), not a certified digital signature.

5.9 Photos and photo pins

Pick **Photo**, click the spot where the photo was taken and pick a picture (IPAD **Take Photo** or **Choose from Library**). A numbered pin marks the spot. Click a pin, with any tool, to see the photo big; on the iPad you can add a **Caption**, **Change Photo** or **Remove Photo**. Photos are added as pages at the end of the exported PDF.

To mark up a photo itself like a drawing, open the JPG, PNG or iPhone HEIC picture as a drawing (2.5). **Export** → **Picture as JPG** sends it back as a picture.

5.10 Snapshot: copy an area as a picture

Snapshot (G , the Snapshot button on the command bar, or right-click the sheet → **Copy area as image…**) copies part of a sheet as a sharp picture, like Bluebeam's Snapshot. Drag a box: the drawing and your markups in it go to the clipboard, ready to paste into Outlook, Word or Microsoft Teams.

- After each box the picture opens in a preview with **Copy**, **Save as PNG…**, **Open** (in Photos), **Paste into drawing** and **Close**. Settings → Appearance → **Show preview after Snapshot** turns the preview off.

- `Ctrl`+`V` on any sheet puts a copied picture there as a picture markup you can move, size and turn. A picture copied from another app works too.
- Snapshot stays on for another box until you press `Esc`.

5.11 Line up, group and stack markups V2

Align and distribute

Select two or more markups, then **Markup** → **Align and distribute** ▸, the **Arrange** buttons in the side panel, or right-click → **Align** ▸:

- **Align left**, **center** or **right** (`Ctrl`+`Shift`+`L` / `E` / `R`), and **Align top**, **middle** or **bottom** (`Ctrl`+`Shift`+`T` / `M` / `B`).
- **Distribute horizontally** or **vertically** (three or more): the outer two stay put and the gaps between the others become equal.

Group

Select two or more and press `Ctrl`+`G` (**Markup** → **Group**, the **Group** button, or right-click → Group). A click on any member then selects the whole group, and moving, copying, Duplicate, Delete and Align act on all of it. Double-click (or `Alt`+click) one member to change it alone; `Esc` selects the whole group again. `Ctrl`+`Shift`+`G` ungroups. IPAD long-press a selection for **Align and distribute** and **Group** / **Ungroup**.

Bring to front and send to back

Right-click any markup → **Bring to front** or **Send to back** (**Edit** menu on Windows, long-press on the iPad). Use it to put a white-out back under your notes, or a note over a cloud.

Rows of symbols

When you place or drag a symbol, stamp or count mark near others of its kind, it snaps into their row: dashed guides show it lining up with a neighbor's center, or one more step at the same spacing. Hold `Alt` to place it freely, or turn off **Snap to markups** in the Symbols, Stamps or Count options.

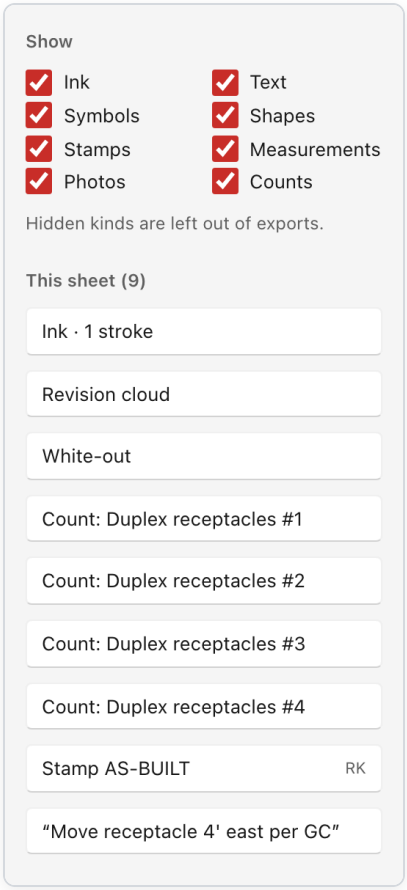
5.12 The Markups list and showing or hiding kinds

The **Markups** tab in the side panel (IPAD the **Markups** button) lists every markup on the sheet: click one to zoom to it.

Under **Show**, untick a kind to hide it: **Ink**, **Text**, **Symbols**, **Shapes**, **Stamps**, **Measurements**, **Photos**, **Counts**. This is how you work in layers on your own drawings: for example hide the counts while you redline.

> **HIDDEN KINDS ARE LEFT OUT OF EXPORTS**
> Show them again before you send the drawing. The iPad asks: **Export without them** or **Show all and export**.

On a Crew sheet there are also **Layers** per person (12.8).



The Markups tab.

6 Symbols and palettes

Draw what's really there with the right symbol: pick it in the panel and click the sheet. Symbols are counted by trade for the count page and the counts spreadsheet.

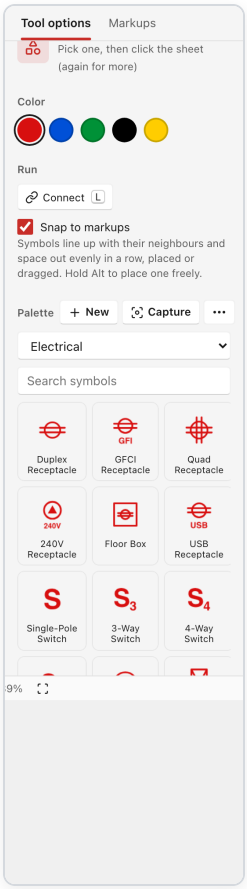
6.1 Place symbols

1. Pick **Symbols** (S).
2. Choose a palette or library in the list (for example **Electrical**), or type in **Search symbols**.
3. Click a symbol, then click the sheet to place it. Keep clicking to place as many as you need. IPAD you can also drag a symbol from the palette onto the sheet.
4. Pick the color first if you want the symbol in another color.

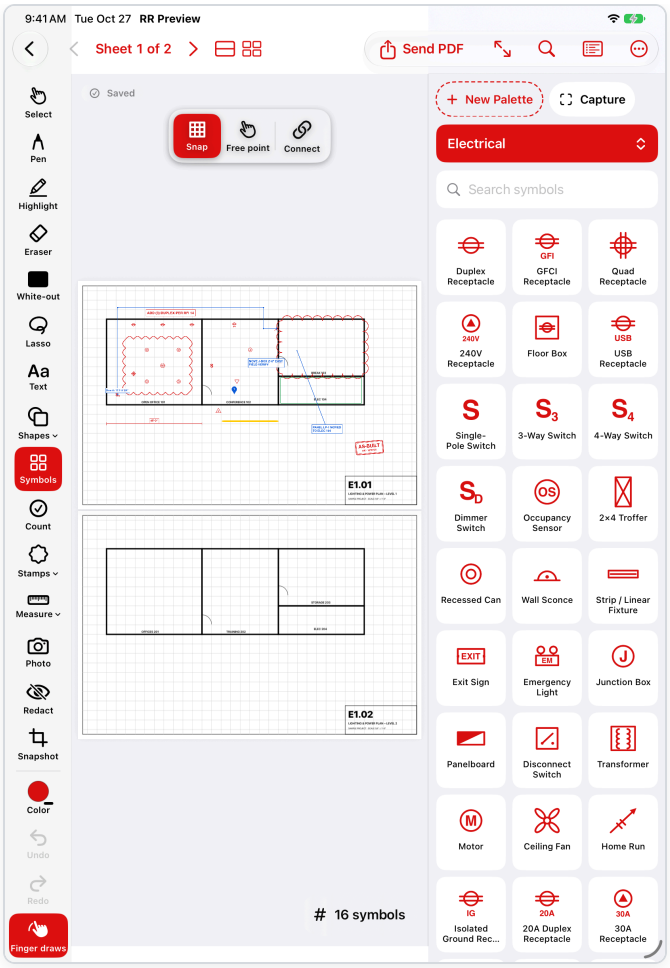
Select a placed symbol to move it, size it with the corner handle, or turn it with the orange dot. **Copy & Place** drops copies at the same size and angle.

Favourites and **Recent** are at the top of the list. Right-click a symbol → **Add to Favourites** (IPAD long-press → **Add to Favorites**).

Tag (in the General library) adds an abbreviation such as DW or MW.



The Symbols panel WINDOWS



The symbol palette on the iPad.

6.2 The symbol libraries

There are 18 libraries. Choose which ones show in **Settings → Symbol libraries**; search only covers the libraries that are switched on. A palette you made keeps its symbols even if their library is switched off.

Library	Symbols	Starts	Library	Symbols	Starts
Contractor Legend	65	off	Gas Piping	20	on
Electrical	110	on	HVAC / Mechanical	47	on
Lighting Controls	21	on	Solar / PV	26	on
Fire Alarm	65	on	EV Charging	20	on
Fire Sprinkler	39	on	Architectural	27	on
Low Voltage	50	on	General	9	on
Telecom / Data	25	on	Medical Gas *(specialty)*	20	off
Security	27	on	Civil / Site *(specialty)*	30	off
Plumbing	50	on	Structural *(specialty)*	20	off

Contractor Legend is a typical contractor's own legend; it stays off until you need it. Security has cameras, card readers and door hardware. General has the revision delta, note bubble, dimension and Tag.

6.3 Connect symbols in a run V2

Show circuiting between devices: in the Symbols panel turn on **Connect** (the chain, or press `L`). Each symbol you place is then joined to the last one with an arc (or a straight line) in your color and line weight. Click a symbol already on the sheet to join it in. `Esc`, `Enter`, a double-click or **Finish run (Esc)** ends the run.

- Pick **Arc** or **Straight**. Right-click an arc → **Flip arc** or **Make straight**, or drag its middle handle to bend it.
- Move a symbol and its arcs follow; delete one and its neighbors are joined.
- The runs spreadsheet lists each run with its symbols and its length when the sheet has a scale.

6.4 My Palettes: your own symbol sets

Your palettes are listed first in the symbol panel. Put the symbols you use on every job in one place:

1. Click **New** (**Tools** → **Palettes** → **New palette…**) and name it.
2. Right-click any symbol → **Add to palette** and pick the palette (or **New palette…**).
3. In a palette, **Edit** lets you drag symbols into order or remove them. ⋯ has **Rename…**, **Save palette…**, **Import a palette…**, **Reorder palettes…** and **Delete**.

Share a palette: **Save palette…** writes a **.redlinepalette** file you can send to another PC or iPad (on the iPad: **Share Palette…** or **Save to Files…**). Opening it brings in the same palette, captured symbols included. If a palette with that name is already there, choose **Replace** or **Keep Both**.

Bluebeam tool chests WINDOWS: **Import a palette…** also takes Bluebeam **.btx** files. Each stamp in the tool chest becomes a captured symbol in a new palette. (The iPad can't import .btx files; import it on a PC and send the .redlinepalette to the iPad.)

6.5 Capture a symbol off the drawing

Need a symbol the libraries don't have, such as one from the legend on the drawing? Copy it:

1. Click **Capture** in the symbol panel (or **Tools** → **Capture Symbol**; iPad: More → **Capture Symbol**).
2. Drag a box around the symbol on the drawing. Only the linework is kept; the white paper is dropped.
3. Name it, pick its palette and, if you like, the heading it's counted under (**Custom** unless you choose a trade).
4. **Use markup color** draws it in your pen color like the other symbols; turn it off to keep the drawing's colors.

A captured symbol is placed, moved, sized, copied and counted like the built-in ones, and goes into every export and .redline file. Right-click it to rename or delete it (sheets that use a deleted one show "?"). To get a symbol from a PDF or a picture, open it as a drawing and capture it; PDFs and pictures can't be imported as palettes.

7 Count and Find & Count

Count receptacles, fixtures or anything else by clicking each one, or let the app find every symbol like the one you box. Totals show in the panel, on the PDF count page and in the counts spreadsheet.

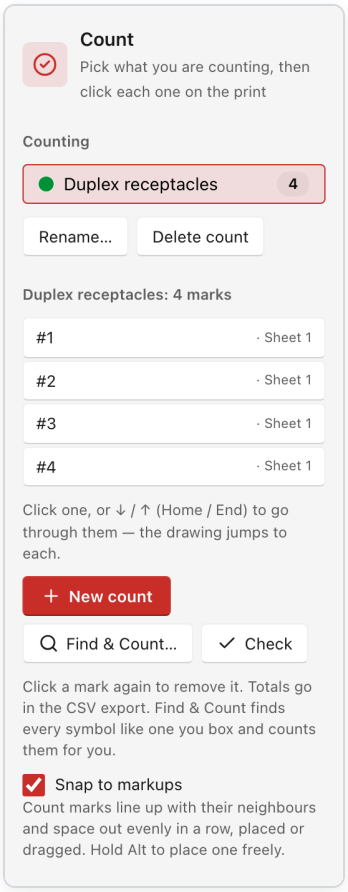
7.1 Count by clicking

1. Pick **Count** (`C`).
2. Click **New count**, type what you're counting (for example *Duplex receptacles*), pick a color and click **Create**. Or click **Check** for plain green check marks.
3. Click each one on the print. Every click drops a numbered check. Click a check again to take it off.

The panel lists your counts with their totals. Pick one to **Rename…** or **Delete count**, and to see its marks on every sheet: click a mark, or press `↓` / `↑` (`Home` / `End`), and the drawing jumps to each one.

Right-click a symbol → **Count under…** starts counting that kind of symbol.

IPAD Count → **New** (with **Quick names**), **Switch** changes the active count, and the bar shows **This sheet** and **All**. Long-press a mark → **Bigger** / **Smaller** changes every mark of that count.



The Count panel WINDOWS

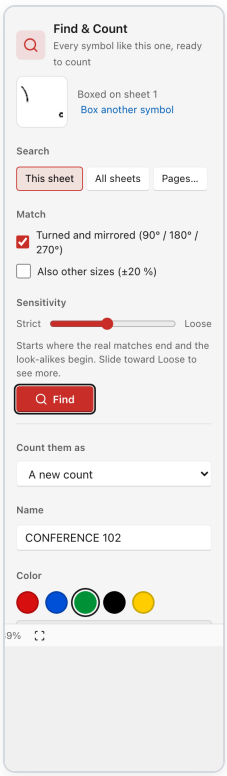
COUNTING TIPS

- Make one count per device type, and work in a pattern (room by room, or left to right) so you don't miss any.
- Watch for look-alikes: duplex, GFCI and quad receptacles look much the same at a glance.
- Recount on every revision. Compare (10.6) shows you where things changed.

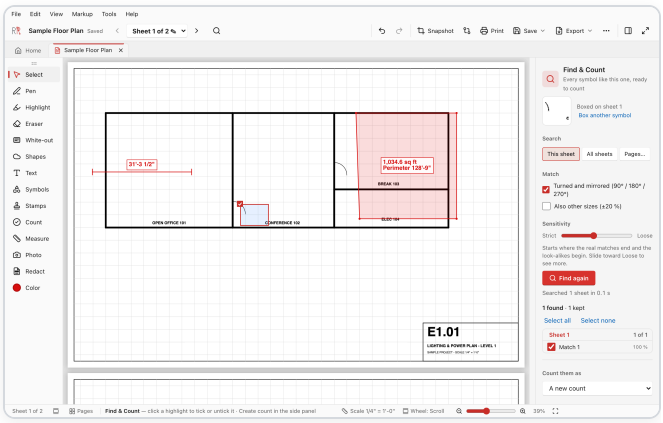
7.2 Find & Count: let the app count a symbol

Find & Count works like Bluebeam's VisualSearch: box one symbol printed on the drawing, and it finds every one like it, on this sheet or the whole set.

1. Start it: **Tools** → **Find & Count…** (`Ctrl` + `Shift` + `F`), **Find & Count…** in the Count panel, or right-click the sheet → **Find & count a symbol here…** (which picks the symbol under the pointer). IPAD More → **Find & Count…**, or long-press an empty spot.
2. **Box one symbol**: drag a box around one symbol printed on the drawing (a receptacle, a fixture, a smoke detector).
3. Pick where to look: **This sheet**, **All sheets** or **Pages…** (for example *1-5, 8*). **Turned and mirrored (90° / 180° / 270°)** is on; tick **Also other sizes (±20 %)** for ones drawn a little bigger or smaller.
4. Click **Find**. The sheets are searched one after another ("Sheet 12 of 172 · 37 found"); keep working, or press **Cancel** (`Esc`).
5. Check the results. Every match gets a box with a tick on the drawing and a row in the side panel, by sheet. Click a row to zoom to it, or step through with `↓` / `↑` (`Space` ticks or unticks). Untick the wrong ones, or click their box on the drawing. **Select all** / **Select none**.
6. Slide **Sensitivity** from **Strict** (only near-perfect copies) to **Loose** (also faint, broken or crowded ones). The list changes as you slide, without searching again; ones you unticked stay unticked.
7. Under **Count them as** choose **A new count** (named after the text next to the symbol when you boxed it in a legend; change the name and color if you like) or a count you already have. Click **Create count**.



Find & Count after boxing a symbol.



Matches ticked on the drawing.

The kept matches become ordinary numbered count marks: totals show in the Count panel, the counts CSV and the PDF count page, also on the iPad. Spots already in that count are skipped. One `Ctrl` + `Z` takes them all back.

GETTING GOOD RESULTS

- Box tightly around **one** symbol, with as little else in the box as you can.
- Box a symbol on the plan, not in the legend: legends are often drawn at another size.
- If you see *No linework in that box — drag around one symbol on the drawing*, the box was on empty paper or text only. Try again around the symbol's lines. *No single symbol right there* means the box needs to go around just the one symbol.
- It works best on vector PDFs (exported from CAD). Scans work when they are clean; check the results, or count by hand.
- The drawing is searched as you see it, including markups already in the PDF (such as Bluebeam symbols), but never your own markups. Signatures are never counted.
- If it stops with *Find & Count stopped: …*, **Copy details** puts the reason on the clipboard for support.

7.3 Where the totals go

- **The Count panel**: per count, with every mark listed by sheet.
- **The count page**: a page listing symbol counts by trade and measured runs at the end of a flattened PDF (Settings → Export → **Include symbol count page**), and the **Counts summary page** option when you print.
- **Counts (CSV)**: a spreadsheet for Excel and the estimate (11.4).
- **Bluebeam**: count marks arrive as grouped Count measurements with totals.

8 Measure

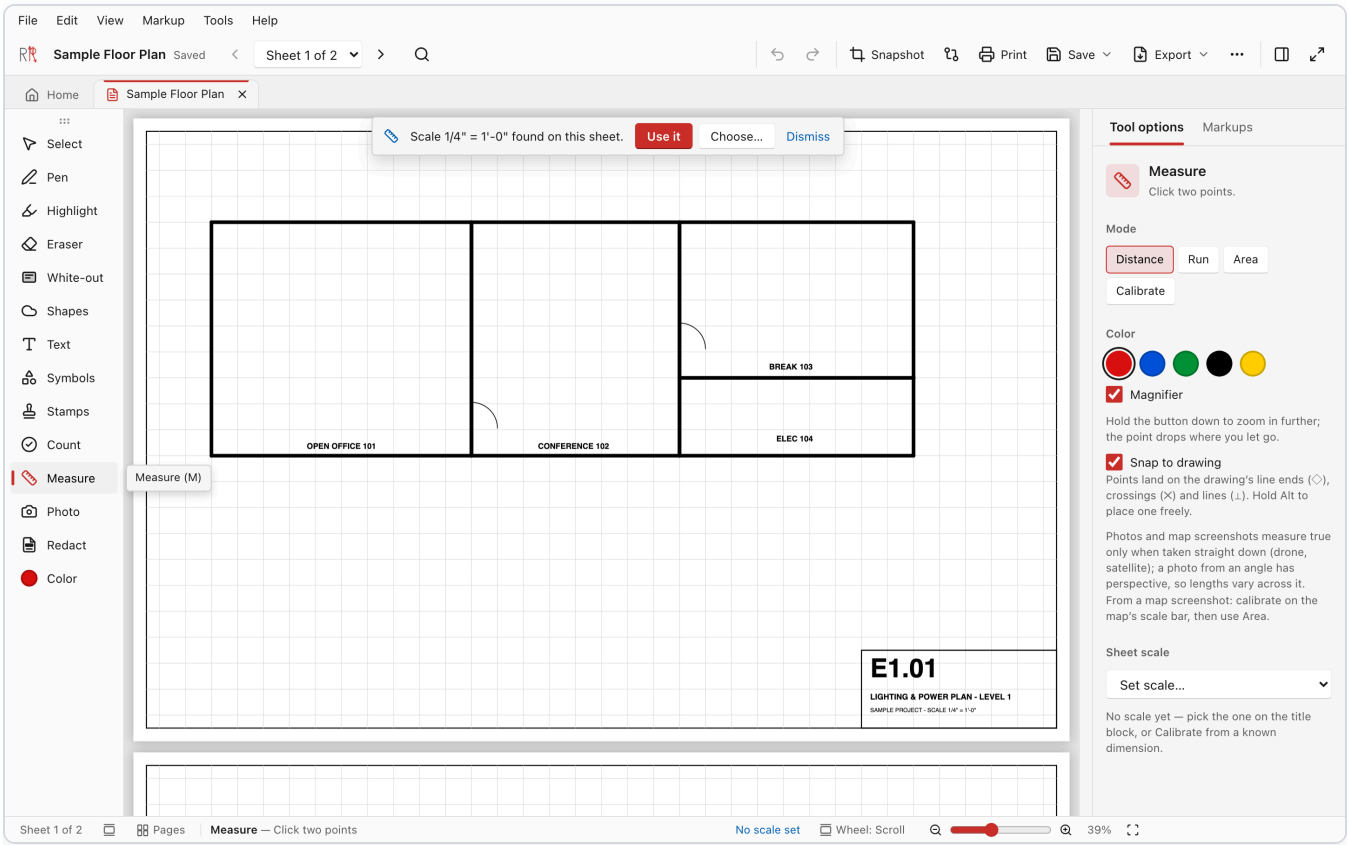
Set the sheet's scale once, then measure distances, runs and areas right off the drawing. Each sheet has its own scale.

8.1 Set the scale

When you pick **Measure** (`M`) on a sheet that has no scale yet, the app reads the scale notes printed on it (1/4" = 1'-0", 1" = 20', 1:100; the title block's is the sheet's scale). A bar over the drawing offers what it found:

- **Use it** sets that scale.
- **Choose…** lists every scale found, by view ("Floor plan – 1/8" = 1'-0"", "Detail 3 – 1/4" = 1'-0"").
- **Dismiss**: it won't ask again on that sheet.

Nothing is set until you click. After **Use it**, if other sheets of the drawing have no scale, the bar offers **Use for all sheets without a scale** (one Undo takes it back) or **Not now**. Just opening a drawing or flipping through it never brings the bar up; only measuring does.



The scale found on the sample sheet, offered in the bar over the drawing.

Choose a scale yourself

Tools → Set scale…, click **No scale set** in the status bar, or use **Sheet scale** in the Measure options. The list shows what was found on the sheet, then the standard scales: Architectural (1/8", 3/16", 1/4", 3/8", 1/2", 1" = 1'-0"), Engineering (1" = 10', 20', 30', 40', 50', 100') and Metric (1:50, 1:100). **Clear scale** removes it. **Precision** rounds lengths to 1/4", 1/8" or 1/16".

Calibrate from a known dimension

When the sheet says NTS, was printed at another size, or is a photo or map: pick **Calibrate**, click both ends of something whose length you know, and type it the way you'd say it: *50 ft*, *10 yd*, *12'-6"*, *30 in* or *3.5 m*. Click **Set scale**.

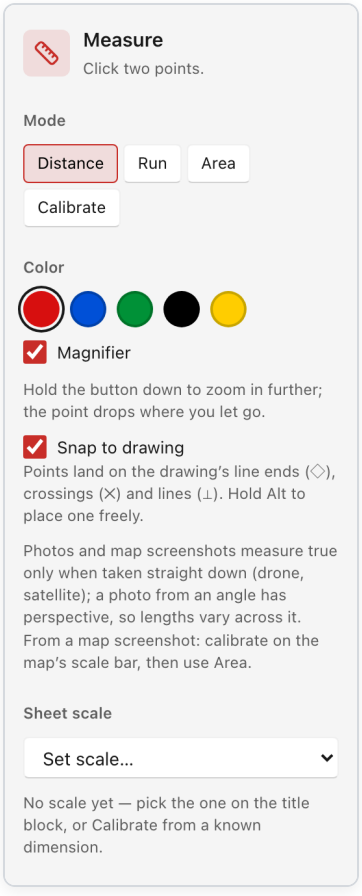
> **CHECK REDUCED PRINTS**
> A half-size print (an 11×17 page whose title block says 22×34, or marked HALF SIZE) offers the halved scale. If the sheet has a graphic scale bar, measure it to check. A scanned sheet with no text is read from its picture by Windows; please verify those. One scale per sheet: for a detail drawn at another scale, calibrate while you measure that detail.

8.2 Distance, run and area

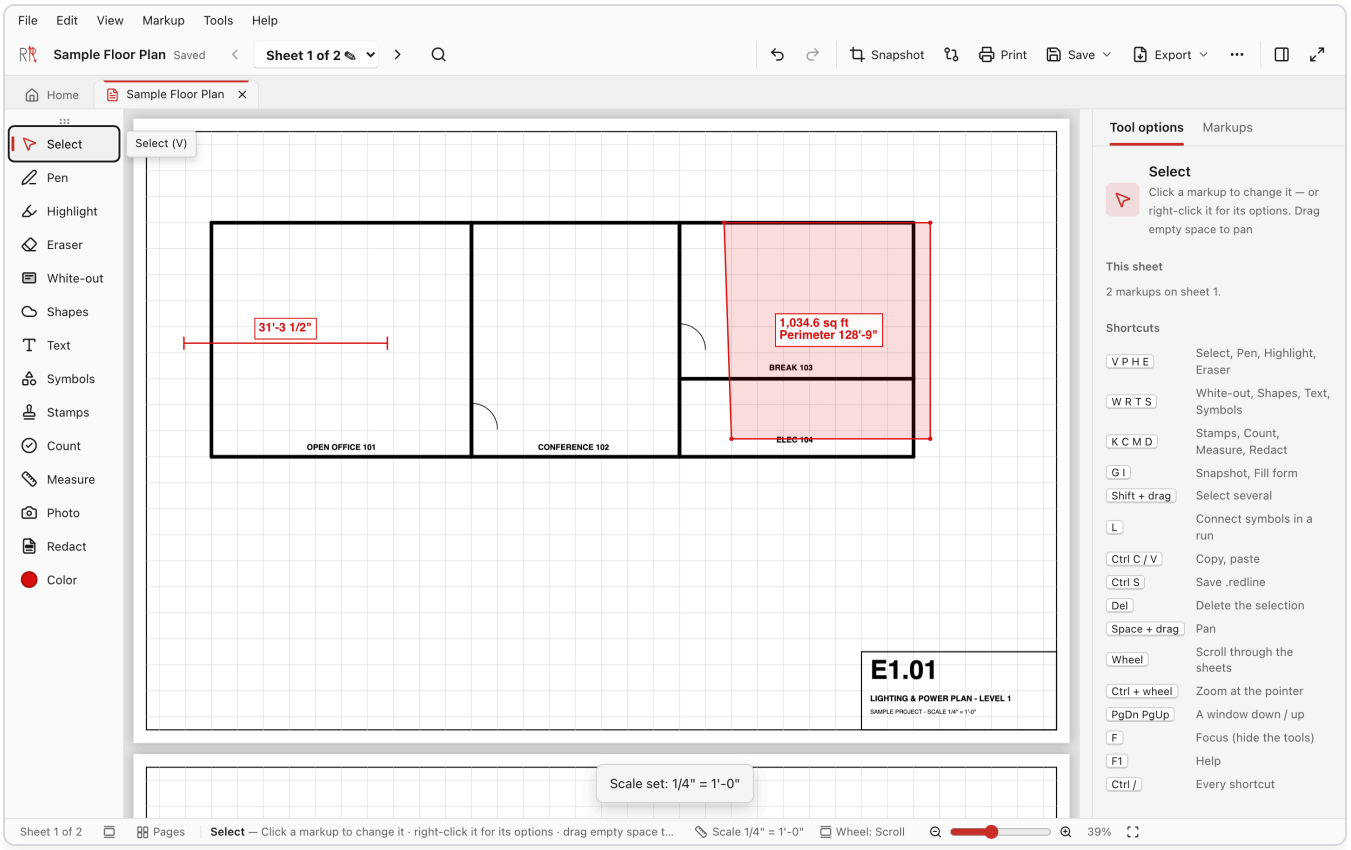
Pick the **Mode** in the Measure options:

- **Distance**: click two points.
- **Run**: click each point, then double-click, press `Enter` or click **Done**. Runs are named and totalled in exports: a conduit or cable run from the panel to the last device.
- **Area** V2: click each corner, then click the first corner again, double-click or press `Enter`. The shape fills lightly and shows its square feet (m² on a metric scale) and perimeter. Drag its corner handles to change it.

IPAD Measure ▾ has **Distance**, **Run (polyline)**, **Area**, **Calibrate Scale** and **Set Scale…**. Tap the last point again or **Done** to finish a run. A selected measurement has **Add Point** and **Remove Point**.



Measure options WINDOWS



A distance across the open office and an area around the break room, at 1/4" = 1'-0".

8.3 Land each point exactly

- **Magnifier**: shows exactly where a point will land. Hold the button down to zoom in further; the point drops where you let go.
- **Snap to drawing** V2: points land on the drawing's line ends (◇), crossings (X) and lines (⊥). Hold `Alt` to place one freely. IPAD **Snap to drawing**, **Snap to markups** or **Free point**; Settings → Measure → **Snap measurements**.
- To fix a point, select the measurement and drag the round handle on its end.

8.4 Photos and maps

A photo measures true only when it's taken straight down (a drone or satellite picture); one taken at an angle has perspective, so lengths change across it. To measure from a map screenshot: open the picture, calibrate on the map's scale bar, then use Area.

9 Forms, redaction and finding text

Fill in a pre-task plan or a permit, take a name or a price out of a drawing for good before it goes out, and find any word on every sheet.

9.1 Fill a PDF form

A PDF with fillable fields (a pre-task plan, a permit, a checklist) gets a **Fill form** button at the top (IPAD **Fill** on the toolbar). Nothing pops up when it opens.

1. Click **Fill form** (or **Markup** → **Fill form**, `I`).
2. Click a field and type. `Tab` and `Shift`+`Tab` go to the next and previous field, on every sheet, and select what's already in it so typing replaces it. `Space` or a click ticks a box or picks a round button; drop-downs open with a click. `Enter` is done with the field; `Esc` takes back what you typed in it.
3. A signature field takes your saved signature, sized into the box. No signature yet? The signature window opens.

The side panel counts the fields ("12 fields · 5 filled in"). The values are saved with the drawing, Undo covers them, and they go into every export and print. When it's done:

- **PDF with the form filled in (fields stay editable)**: the PDF for Bluebeam / Acrobat.
- **Flatten form — burn the values in so it can't be changed**: **Export** → **Filled form, flattened…**. Your markups stay editable. IPAD the Send PDF menu has **Flatten form (Bluebeam / Acrobat PDF)**.
- **Clear form** empties it.

With any other tool, a click on a field belongs to that tool, so you can mark up the form as usual.

> **ANSWER TOO LONG?**
> A field turns amber when its answer is too long for the box. The exported PDF writes it smaller so all of it shows, down to 6 pt; an answer too long even then is cut off at the end, so shorten it.

Add forms… CREW V2: in a local project or a Crew project, **Add forms…** adds fillable PDFs as documents: each file stays one document with all its pages and its fields fillable (10.7, 12.6).

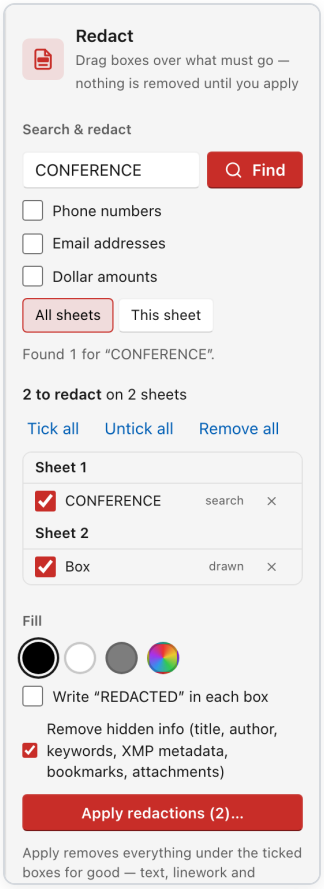
9.2 Redact: remove things for good

Redact removes things from a drawing for good (a name, a phone number, a price, part of a picture) before you send it out. White-out only covers; Redact takes it out. It works in two steps: first you **mark** what must go, then you **apply**.

Step 1: mark what must go

1. Pick **Redact** (D, or **Tools** → **Redact**).
2. Drag boxes over what must go, on as many sheets as you like. They show as red hatched boxes until you apply them. Click a box to untick it (it stays but won't be redacted); right-click it to remove it.
3. Or let it find text for you: under **Search & redact** type a name, word or number, or tick **Phone numbers**, **Email addresses** or **Dollar amounts**. Pick **All sheets** or **This sheet** and click **Find**. Every match becomes a box you can untick in the list (click a row to see it).
4. Choose the **Fill** (Black, White, Gray or any color), whether to **Write "REDACTED" in each box**, and **Remove hidden info** (title, author, keywords, XMP metadata, bookmarks, attachments).

The marked boxes are saved with the drawing, so you can close it and finish another day. Nothing is removed until you apply.

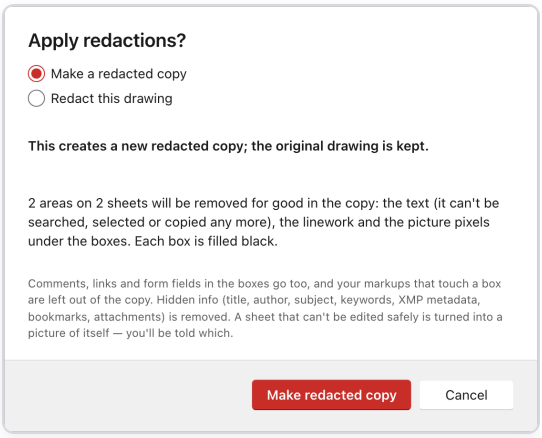


Two boxes marked: one found by Search & redact, one drawn.

Step 2: apply the redactions

Click **Apply redactions…** (**Tools** → **Apply redactions…**). It asks every time:

- **Make a redacted copy**: a new drawing, "*name* (Redacted)", opens in a new tab; the original is kept as it was. If there is a redacted copy already, pick **Update existing copy** or **New copy** ("(Redacted 2)").
- **Redact this drawing**: removes the content from the open drawing itself. Undo puts it back until you close its tab; after that it is permanent.



Apply redactions?

Everything inside the boxes is removed from the PDF itself: the text (it can no longer be searched, selected or copied, and the words next to a box keep their place), the linework, the picture pixels, and comments, links and form fields in the boxes. Your own markups that touch a box are left out. If a sheet can't be edited

safely (an unusual font or picture format), it's turned into a picture of itself at up to 300 dpi with the boxes painted in, and you are told which sheets.

> **CHECK BEFORE YOU SEND**
> Open the redacted copy in Acrobat or Bluebeam and try to search for and select what you removed.

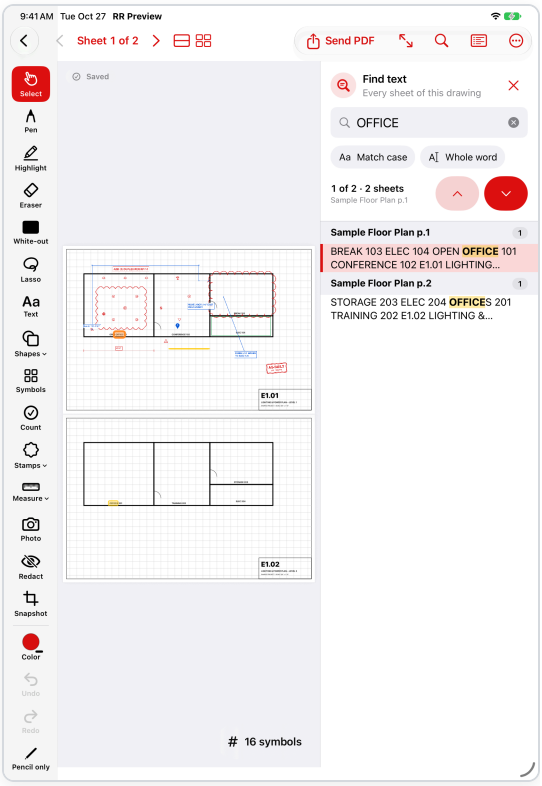
9.3 Find text on every sheet

Find any word printed on the drawing: a room name, MILLWORK, a panel number, a note.

1. **Edit → Find…** or `Ctrl` + `F` (the magnifier on the command bar; IPAD **Find** in the top bar).
2. Type. Every place it is printed is highlighted, the one you're on in orange.
3. `Enter` (or › and `F3`) goes to the next one, `Shift` + `Enter` to the one before, from sheet to sheet.

The side panel's **Search** tab lists the hits by sheet with the words around each: click one to go there. Capitals don't matter unless you turn on **Match case**; **Whole word** finds DOOR but not DOORS. A hyphen or line break is ignored, so *millwork* finds MILL-WORK split over two lines. `Esc` closes Find.

Scanned sheets are pictures with no text in them, so they can't be searched; ask the office for a PDF exported from CAD or Revit.



Find on the iPad.

10 Sheets and drawing sets

Get around a big set, put a set together, compare a new revision with the old one, and keep drawings together by job.

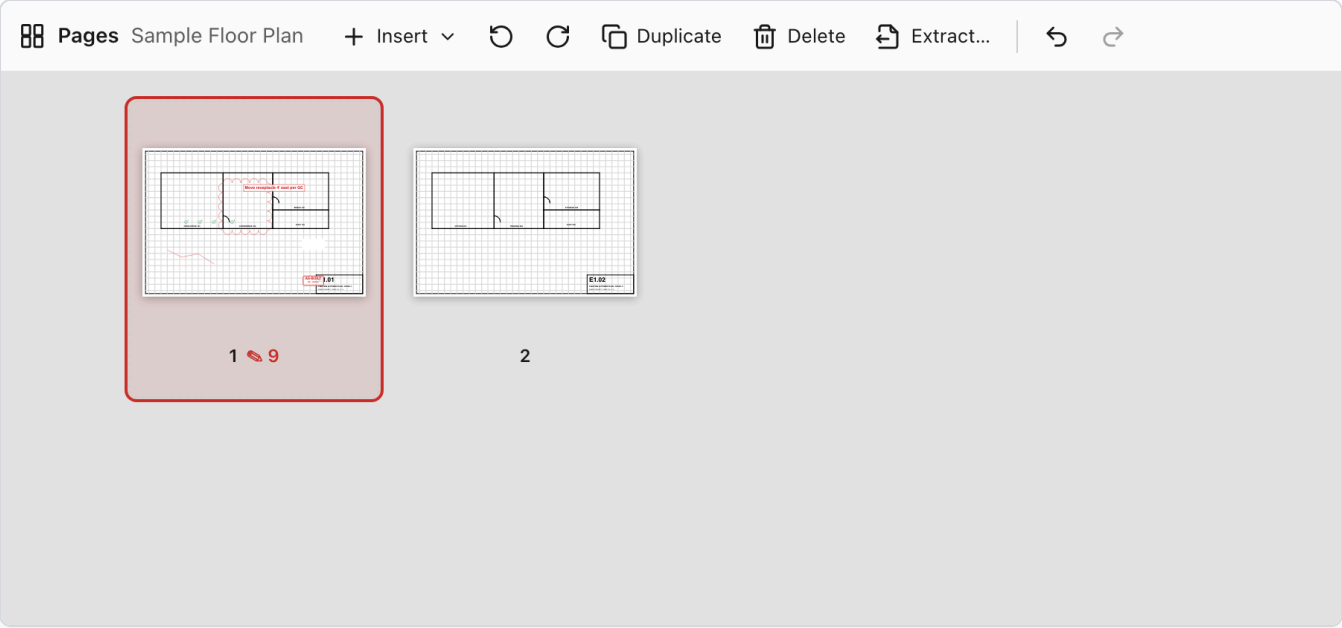
10.1 Sheet names and going to a sheet

When the PDF has bookmarks or page labels, the app uses them as sheet names: "G101 · First Floor Code Plan — 12 of 130". Otherwise it shows "Sheet 12 of 130".

- WINDOWS Click the sheet name on the command bar or in the status bar for the list, or **View → Go to sheet…** and type a sheet number, its title or its position (`Ctrl`+`G` when nothing is selected to group). `Page Down` / `Page Up` go to the next and previous sheet.
- IPAD Tap **Sheet n of N** for the **Sheets** list with thumbnails, or use ‹ ›.
- **Links**: links in the PDF work with Select, and with Settings → Links → **Link sheet references** on, sheet numbers written on the drawing (3/E5.01, SEE E6.01) work as links too. On the iPad, Back returns.

10.2 Pages: reorder, rotate, insert and extract

Put a set together without Acrobat. **View → Pages…** (`Ctrl`+`Shift`+`P`, or **Pages** in the status bar; IPAD **Pages** next to Sheet n of N) shows every page as a thumbnail with its markups.



The Pages view with its buttons along the top.

- **Select**: click a page; `Ctrl`+click adds or removes one, `Shift`+click picks a run, `Ctrl`+`A` picks them all. IPAD tap pages, or **Select All** / **Select None**.
- **Move**: drag pages; a line shows where they will go.

- **Rotate** 90° left or right, **Duplicate** (`Ctrl`+`D`), **Delete** (`Del`), or **Extract…** to a new drawing that opens in its own tab (tick **Then delete them from this drawing** to move them out).
- **Insert** ▸ before or after the selected page, or at the beginning or end:
  - **PDF pages…**: all of another PDF, or a range such as *1-3, 5*.
  - **Pictures…**: JPG, PNG or iPhone HEIC, one per page, on Letter, 11×17, A1–A4 or the size of the drawing's first page, with an optional caption (the file name and/or the date the photo was taken).
  - **Blank page…**: the size of the selected page, Letter, 11×17, Arch D or A1–A4.
  - **Word or Excel…** WINDOWS: see below.
  - **Notes page…** V2: see 10.5.
- Double-click a page (or press `Enter`) to go to it; **Done** or `Esc` closes the view.

Your markups stay on their pages: moved, turned with the sheet, copied with a duplicate, removed with a deleted page. Each change is one Undo step. Pages are copied as they are (still sharp vector drawings with their text and links), so exports, Print, Redact and Compare all get the new order.

To add pages without opening Pages: **File** → **Insert PDF pages…** or **File** → **Insert pictures…** (iPad: More → **Insert PDF pages…** / **Insert pictures…**), or drag PDFs or pictures onto the open drawing and choose **Insert into this drawing**.

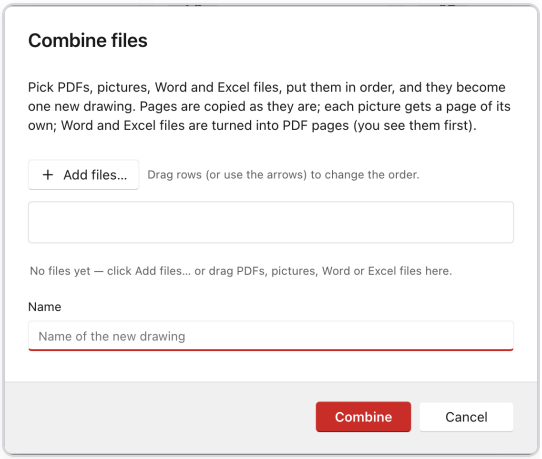
> **PROTECTED PDFS**
> A protected (encrypted) PDF can't be rearranged. Save an unprotected copy first.

10.3 Combine files into one drawing

Click **Combine files** on the Home tab (or **File** → **Combine files…**, or drag several PDFs onto the Home tab). Add PDFs and pictures, put them in order with the arrows or by dragging, name the new drawing and click **Combine**. The result counts as one drawing.

WINDOWS Combine files, Insert ▸ Word or Excel… and **File** → **Insert Word or Excel…** also take Word (.docx, .doc, .rtf) and Excel (.xlsx, .xls, .csv) files. Each one is turned into PDF pages and shown first. If Microsoft Word or Excel is installed, it makes the pages exactly as Save as PDF would ("Uses Microsoft Word on this PC"); without Office, the app converts them itself ("Simplified conversion"). For a workbook, pick **All sheets with content**, the **Active sheet only**, or tick the sheets you want.

ANDROID Combine files takes PDFs and pictures only; save Word and Excel files as PDF first.



Combine files WINDOWS

10.4 Paper sizes

The app knows the US sizes (Letter, Tabloid / 11×17, ANSI A–E, ARCH A–E) and the metric A sizes. They are used for blank sheets, inserted pictures and blank pages, printing, and to check the scale of a sheet that was printed at another size.

Where	Sizes
Blank sheet WINDOWS	Letter 8.5×11, Tabloid 11×17, Arch D 24×36, Arch E 36×48, A1, A2, A3, A4
Blank sheet IPAD	Letter 8½ × 11, Tabloid 11 × 17, Arch D 24 × 36, Arch E 36 × 48
Insert pictures / blank page	Letter, 11×17, Arch D (blank page), A1–A4, or the drawing's own size
Print paper	Same as drawing, ANSI A–E, ARCH A–E1 WINDOWS also A4, A3, A2, A1

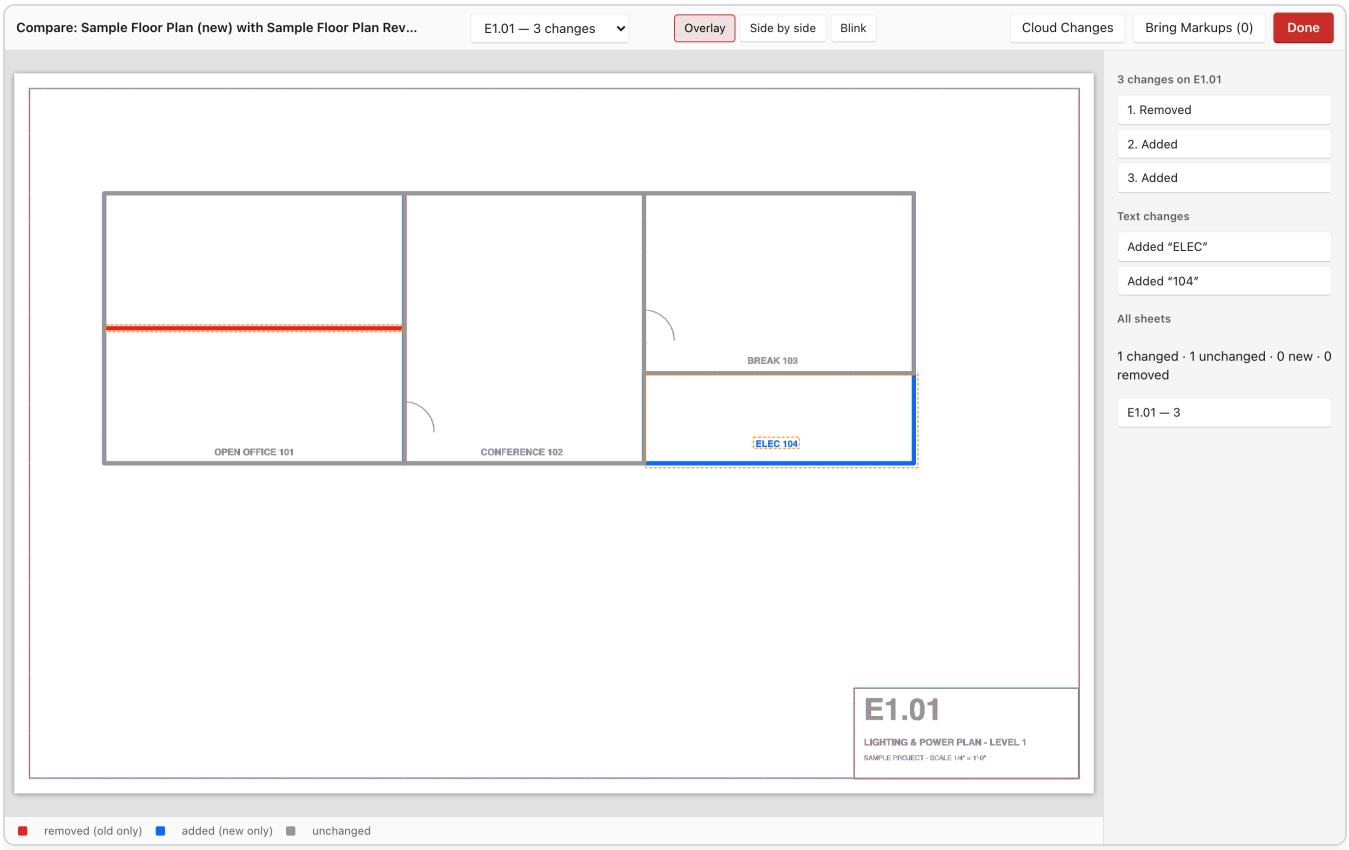
10.5 Notes pages V2

A notes page is an editable page of field notes added to the drawing: **File** → **Add notes page…**, the ⋯ menu, right-click the sheet, or Insert ▸ in Pages (iPad: More → **Add notes page…**).

- A **Title** (for example "Field Notes — E2.03"), a **Numbered list** with a Sheet / grid column, and **Free text** above or below the list.
- **Fill from markups** (or **Fill from all markups**) collects the drawing's text notes, callouts and text stamps into the list. On the iPad, **Place numbered bubbles on the sheets** links them to the drawing.
- **Page size**: Same as this sheet, or Letter. Put it after this sheet or at the end. Long lists flow onto more pages; author and date go in the footer.
- Double-click the page (or **Edit notes**) to change it.

10.6 Compare a new revision with the old one

Got a new revision? Open the new print, then **Tools** → **Compare with older print…** (the Compare button on the command bar; iPad: More → **Compare with Older Print…**, or long-press the drawing on the home screen). Pick the older print on this device, or **From a file…**, and click **Compare**. Sheets are matched by sheet number (E2.04 with E2.04), so added or reordered pages still line up.



Comparing the sample drawing with its older revision: removed in red, added in blue.

- **Overlay** shows what was removed in red and what was added in blue; unchanged lines are grey.
- **Side by side** zooms both prints together. **Flip** / **Blink** flips between old and new in the same spot (Space).
- The change list shows each changed area: click one to zoom to it. The summary shows which sheets changed, were added or removed.
- **Cloud Changes** draws revision clouds around the changes on the new print (Undo removes them).
- **Bring Markups** copies your markups from the old print onto the new one and tells you which landed on changed areas.

10.7 Projects on this device V2

Keep drawings together by job. Projects on this device stay on this PC or iPad: nothing is uploaded and no Crew sign-in is needed.

1. On the home screen click **New project**. Type the **Project name** (for example "Tower Medical Center") and, if you like, the **Project number** (for example 24-118). Click **Create**.
2. Drag drawings onto the project's card, or right-click a drawing → **Move to project…** (iPad: **Move to project…**).
3. Open the project for **Add drawings…** (PDFs, pictures, Word / Excel or .redline files; each file is one drawing) and **Add forms…** (fillable PDFs).

The project's menu has **Open**, **Add drawings to this project…**, **Rename…** and **Delete project….** Deleting a project puts its drawings back under **Drawings not in a project**; **Delete with N drawings…** removes them too (it asks again). **Search projects and drawings** looks at drawing names, project names and project numbers. Every move can be undone from its message.

Free drawings are still counted per drawing, not per project.

11 Sending your work

Your markups always stay editable in the app. When you send, you choose what the office gets: a PDF they can keep editing in Bluebeam or Acrobat, a flattened PDF for the record, a print, a spreadsheet of counts, or the whole job as a .redline file.

11.1 Editable or flattened?

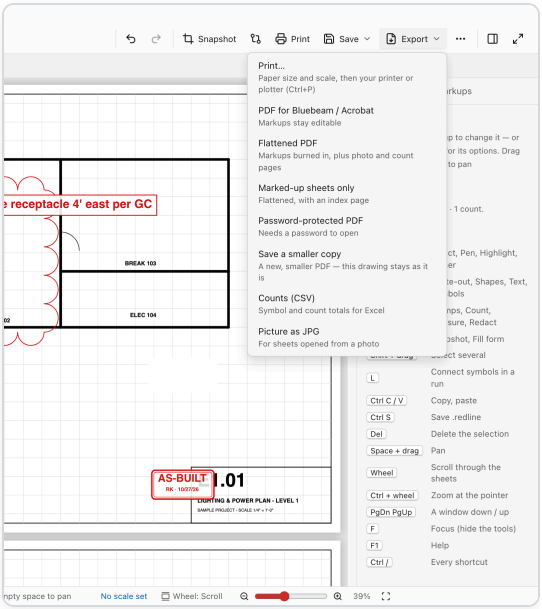
	PDF for Bluebeam / Acrobat	Flattened PDF
Markups	Real PDF markups: each one listed in Bluebeam's or Acrobat's Markups List with Author (your initials), Subject (such as Duplex Receptacle, Revision Cloud) and Date, and can be moved, edited or deleted there.	Burned into the sheets. Nobody can move or remove them.
Symbols	Drawn shapes; captured symbols and signatures as stamps.	Part of the drawing.
Measurements and counts	Live Bluebeam measurements at the sheet's scale; count marks as grouped Count measurements with totals.	Burned in, plus the count page.
Photos	—	Added as pages at the end.
Use it for	Sending to a PM who works in Bluebeam or Acrobat.	The record copy; anyone who just needs to read it.

Markups a PM adds in Bluebeam show on the sheet when the PDF comes back. Opening a PDF that was exported from the app (also after Bluebeam or Acrobat edited it) offers its markups back as editable markups.

11.2 Export WINDOWS

The **Export** menu on the command bar (or **File → Export**):

- **Print…**: paper size and scale, then your printer or plotter (`Ctrl` + `P`). See 11.6.
- **PDF for Bluebeam / Acrobat**: markups stay editable.
- **Flattened PDF**: markups burned in, plus photo and count pages.
- **Marked-up sheets only**: flattened, just the pages with markups, with an index page.
- **Password-protected PDF**: needs a password to open (below).
- **Save a smaller copy**: a new, smaller PDF; this drawing stays as it is (below).
- **Filled form, flattened** (9.1).
- **Counts (CSV)**: symbol and count totals for Excel (11.4).
- **Picture as JPG**: for sheets opened from a photo.



The Export menu.

You choose where each file goes in the usual Save dialog. There is no email button on Windows: attach the file in Outlook, or paste a Snapshot.

Password-protected PDF

Pick which PDF (for Bluebeam / Acrobat, Flattened, or Marked-up sheets only), type the **Password to open it** twice, and choose **Allow printing**, **Allow copying text** and **Allow editing and comments**. It uses AES-256 and opens in Acrobat, Bluebeam, Edge, Chrome and on phones. There's no way into the file without the password, so keep it somewhere safe.

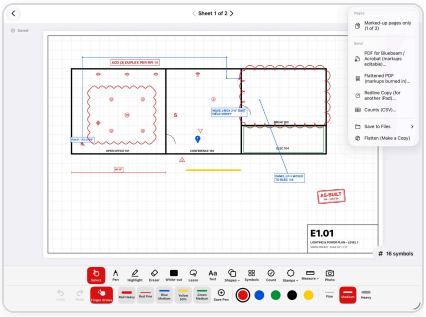
Save a smaller copy

Makes a new file, "*name* (reduced).pdf", for emailing. **Lossless** changes nothing you see; **Also reduce pictures to** 300, 200 or 150 dpi makes scans and photos softer, so use it for emailing, not for the record. Choose whether the markups are flattened or kept as annotations, then **Save copy…**.

11.3 Send PDF IPAD

Tap **Send PDF** in the top bar (**Send** on an iPhone). At the top are the options: **Marked-up pages only (n of N)**, **Include panel schedules** and, on a form, **Flatten form (Bluebeam / Acrobat PDF)**. Then pick what to send:

- **PDF for Bluebeam / Acrobat (markups editable)…**
- **Flattened PDF (markups burned in)…**
- **Redline Copy (for another iPad)…**: a .redline file (11.8).
- **Counts (CSV)…**



The Send PDF menu.

- **Panel Schedules (Excel)…** when the drawing has panels.

The share sheet opens: Mail, Messages, Teams, OneDrive, AirDrop and so on. **Save to Files** ▸ saves instead. **Flatten (Make a Copy)** adds a flattened copy on the home screen, and **Print…** prints. A drawing opened from a photo can be sent as a picture: **Send as Image (JPG)…**.

ANDROID Every export asks **Share…** (the Android share sheet: Gmail, Outlook, Drive, OneDrive, Teams, Bluebeam, a text message) or **Save to…** (a folder on the tablet or in the cloud).

11.4 Counts spreadsheet (CSV)

Counts (CSV) saves the symbol counts and count totals for Excel or your estimate: one row per sheet, trade and symbol, with the columns **Sheet**, **Page**, **Trade**, **Symbol** and **Count**. When the drawing has runs, a second file lists them: **Sheet**, **Page**, **Run name**, **Length (ft)** and **Scale** (and the symbols in each connected run). On a Crew sheet the counts also say who made them.

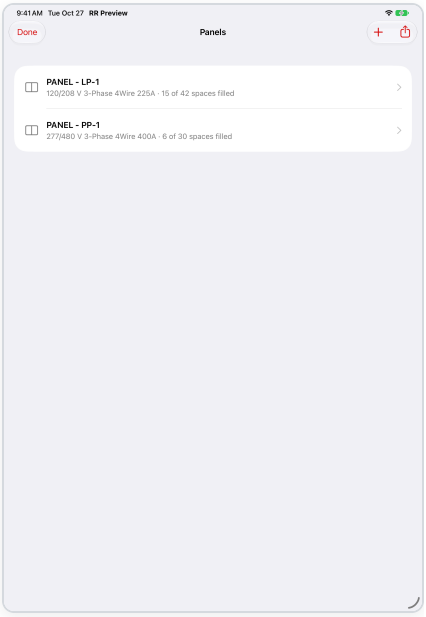
11.5 Panel schedules IPAD

Panel schedules are made on the iPad. Open the drawing, then More → **Panel Schedules…** → **New Panel**:

1. Type the **Panel name** (for example LP-40), the number of spaces, **Voltage**, **Amps** and **Location**.
2. Tap a circuit to fill in its **Description**. **Quick phrases**, **Rooms on this drawing** (for example "Recepts rms B121,123-B126") and **Used on this job** save typing. **Mark Spare** prints it grey.
3. For a 2- or 3-pole breaker: **Select**, pick the circuits, then **Combine (2-Pole)** or **Combine (3-Pole)**. **Split** takes it apart.

Send Panel Schedules… (or **Send This Panel…**) makes an Excel file in the directory-card layout (two panels per page) and the same cards as a PDF, ready to send to the PM. **Open Panel Schedule (.xlsx)…** brings in the office's sheet. Each card is headed with your Settings → **Company profile**.

WINDOWS and ANDROID: **Tools** → **Panel schedules…** shows a sheet's panels read-only and saves them as a PDF. Panels stay in the .redline file when you save it again.

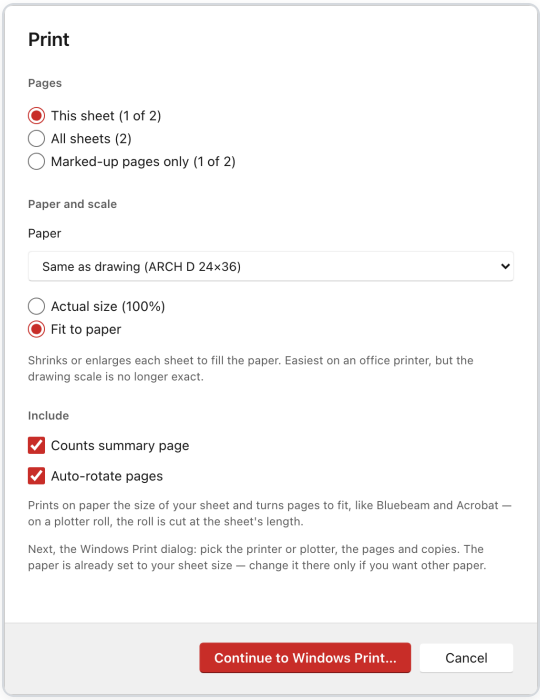


Panels on the sample drawing.

11.6 Print and plot

Click **Print** (`Ctrl` + `P`; iPad: Send PDF → **Print…**). The sheet prints with every markup burned in, sharp at any size.

- **Pages**: This sheet, All sheets, or Marked-up pages only. Tick **Counts summary page** to add the counts for the printed pages.
- **Paper**: **Same as drawing** (the sheet's own size, such as ARCH D 24×36) or a standard size from ANSI A 8.5×11 to ARCH E 36×48.
- **Scale**: **Actual size (100%)** prints true to scale, so a 1/4" = 1'-0" drawing still measures; the printer needs paper as big as the sheet. **Fit to paper** shrinks or enlarges each sheet to fill the paper (no longer to scale). When the paper has the drawing's proportions (24×36 on 12×18) you also get the exact reduction, such as **Half size (50%)**: then 1/4" = 1'-0" prints at exactly 1/8" = 1'-0".
- **Auto-rotate pages** (on) turns pages to suit the paper or the plotter roll.



Print options WINDOWS

WINDOWS Click **Continue to Windows Print…**. The standard Windows Print dialog opens straight away: pick the printer or plotter, the pages and copies, then Print. The paper is already set to your sheet size. **Microsoft Print to PDF** makes a PDF of the print.

> KEEP IT TO SCALE
> Keep any scaling in the printer's settings at 100% (on the iPad, leave **Scaling** at 100% in the print panel). Hidden markups are left out of the print.

IPAD The iPad prints with AirPrint. Plotter not showing up? Many older plotters don't support AirPrint: send the Flattened PDF to the plotter maker's app (HP Smart / HP Click, Canon PRINT), or print from the Windows app. ANDROID **Continue to Print…** opens Android's print screen; printers come through print service apps such as HP Smart or Mopria.

11.7 Snapshot and pictures

For an email or a text, a Snapshot (5.10) of the part that matters is often better than the whole PDF. A drawing opened from a photo exports as a picture with **Export → Picture as JPG** (iPad: **Send as Image (JPG)…**).

11.8 Move work between iPad and PC: .redline files

A **.redline** file holds the whole job: the drawing, every markup, the photos and any captured symbols it uses. Someone with Rough-In Redlines on a PC, an iPad or an Android tablet opens it fully editable; initials show who made each markup. It's also how to back up your work.

1. WINDOWS **Save → .redline file** (`Ctrl`+`S`). After the first time, `Ctrl`+`S` updates the same file. **Save as…** (`Ctrl`+`Shift`+`S`) makes a new one.
2. Send it the way you send any file: email, OneDrive, Teams, a USB stick.
3. IPAD Tap the file in Mail, Teams, OneDrive or Files (or AirDrop it) and it opens in the app. Mark up, then Send PDF → **Redline Copy (for another iPad)…** to send it back.
4. Back on the PC: **Open file** and pick the .redline.

If that drawing is already there, you can **Replace** it or **Keep both**. Markups, counts and photos come along. **Save markups only (.redline.json)…** WINDOWS saves just the markup data, without the drawing.

> **SIGNED IN TO CREW?**
> Crew sheets sync by themselves; you don't need .redline files to share them with your crew (chapter 11).

11.9 Backups

- WINDOWS Drawings are saved in the app as you work, and a backup copy is kept on the PC. If a drawing can't be read, it's put back from that backup and the Home tab tells you. **Reopen drawings from last time** brings back your tabs after a restart or a crash. For a copy off the PC, save .redline files.
- IPAD Settings → **Backup** → **Choose Backup Folder…** and pick a folder in OneDrive, SharePoint, Google Drive, iCloud Drive or Dropbox. Every drawing you change is copied there as an editable .redline file after you stop editing and when you leave the app. If the iPad is lost or broken, **Restore from Backup…** brings the drawings back with markups still editable. Backups are never deleted from the folder, even when you delete a drawing in the app. The home screen shows **Backed up …** or **Backup pending**.
- ANDROID Drawings live inside the app on the tablet, and uninstalling the app removes them. Save .redline files to keep a copy.

12 Working as a crew CREW

Rough-In Crew shares drawings and markups with everyone on the job. The office uploads the set once (or links it from Procore); each person's PC or iPad downloads the sheets so they open without internet; markups sync both ways, live.

12.1 What Crew is

- **Projects** with their drawing sets, shared with the people on them. Each sheet keeps its revisions.
- **Shared, live markups**: everyone's markups on the same sheet, each person on their own layer.
- **Offline**: the sheets you chose are kept on your device; markups made offline sync when you're back online.
- **Items** (a punch list), comments, notifications, **Send to office**, **share links** for people without the app, and an optional **Procore** link.

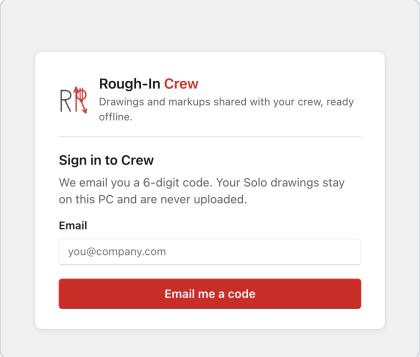
Crew is built into the Version 2 apps on Windows (Rough-In Redlines Preview) and the iPad (RR Preview). Your own drawings (not in Crew) stay on your device and are never uploaded. Drawings uploaded to Crew are kept in **Rough-In Redlines Cloud**.

> PLANS
> Each person who works in Crew needs a personal subscription ($14.99/month through Apple on iPhone and iPad, $15/month on Windows; up to 3 devices) or a license their company buys ($15 per person per month; 1–5 devices). Crew includes Rough-In Pro. A new company can try it free for 7 days with up to 4 people. When a plan or subscription lapses, projects stay there to view but can't be changed; nothing is deleted. Details and how to start: 3.2.

12.2 Sign in

1. Open **Crew**: the Crew tile on the Home tab WINDOWS, or the **Crew** button on the home screen IPAD.
2. Under **Sign in to Crew**, type your work **Email** and click **Email me a code**.
3. Check your email for the 6-digit code, type it under **Code** and click **Sign in**. Wrong address? Click **Change**.

There is no password to remember. Emails used on this device before are listed under **Signed in here before**; click one to fill it in.

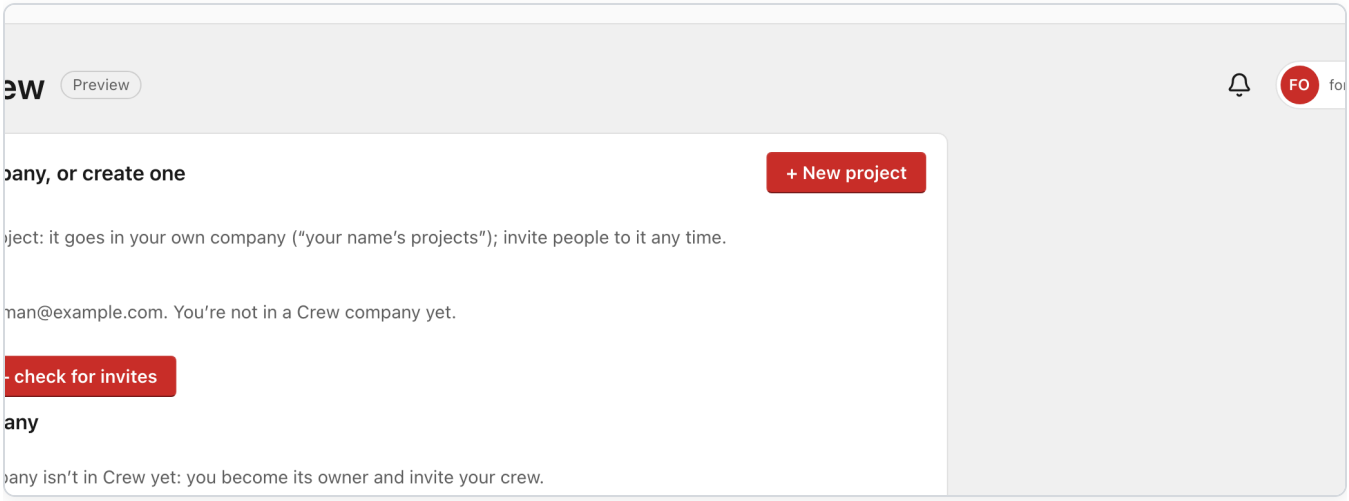


Sign in to Crew WINDOWS

Your company

The first time, you either join your company or create it:

- If someone already invited you, you land in your company. Otherwise click **I was invited — check for invites**.
- If your company has claimed its email domain, signing in with your work email joins it by itself ("Welcome to … Crew"). Public email (gmail.com, outlook.com…) never counts.
- **Create a company**: only if your company isn't in Crew yet. You become its owner and invite your crew.
- Or just start a project with **+ New project**: it goes in your own company, and you can invite people to it any time.



Join your company, or create one.

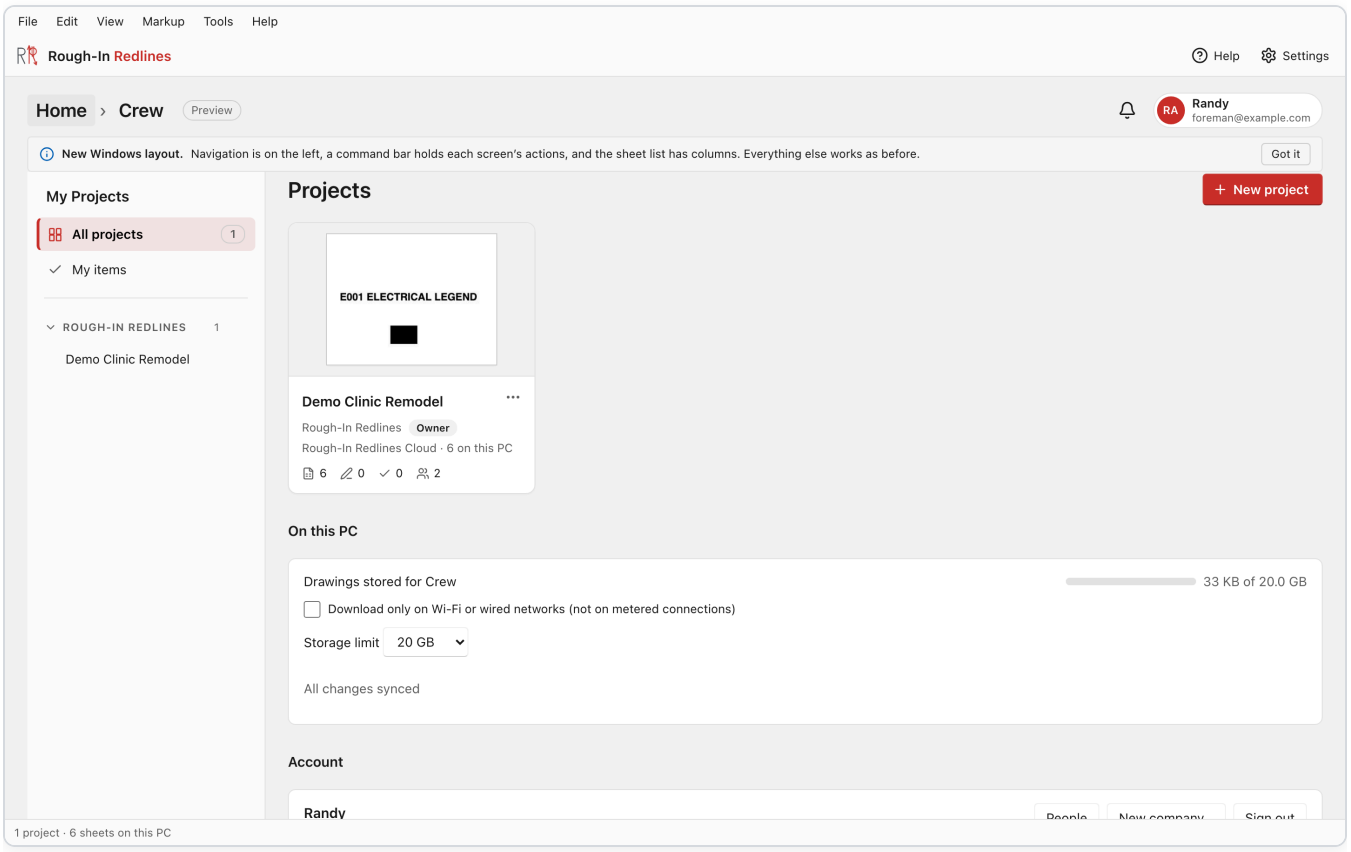
Someone in several companies sees each company as its own section on the Crew home screen.

12.3 Roles: who can do what

Role	Where	What they can do
Owner	Company	Everything. Only an owner can make another owner, set the email domain, or delete a project for good.
Admin	Company	Every project; invites people, links projects, archives and deletes archived projects.
Member	Company	Only the projects they're added to, with a project role:
Manager	Project	The project's people, drawings and settings: rename, project number, hide sheets for everyone, archive.
Editor	Project	Marks up, sends to the office, adds drawings and forms, makes share links.
Viewer	Project	Sees drawings and markups; can't change anything. Sheets open **View only · updates live**. Can comment if the company allows it.

Anyone in a company can start a project and becomes its manager.

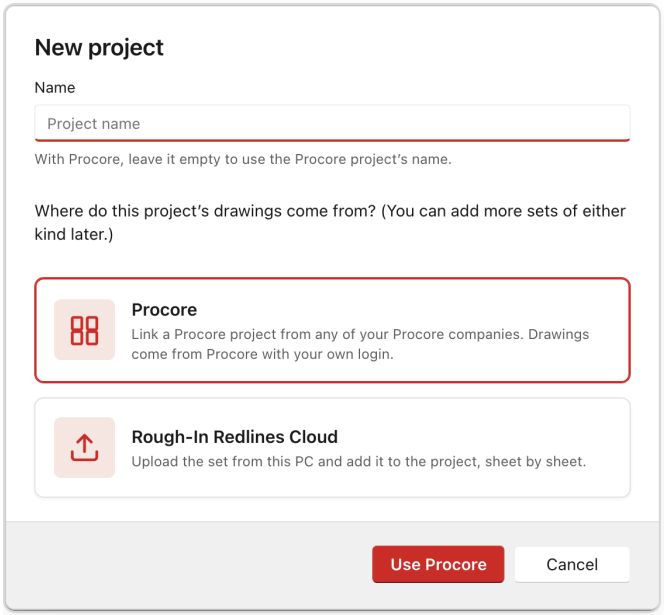
12.4 The Crew home screen and projects



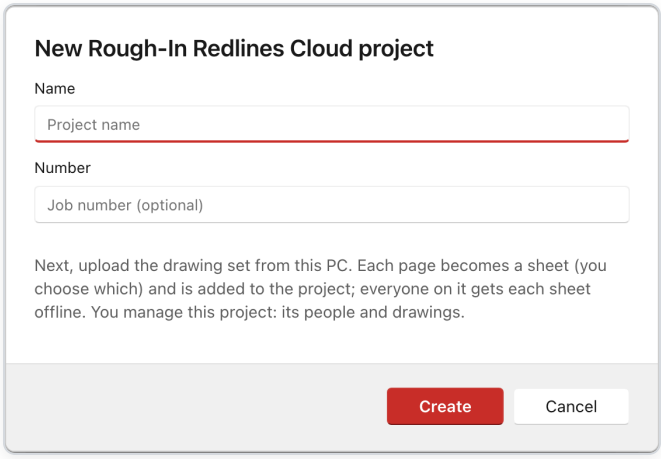
The Crew home screen on Windows: navigation on the left, the projects, and On this PC.

- WINDOWS The left side lists **My Projects**: **All projects**, **My items**, each company and its projects, and **Archived**. The bell at the top opens **Notifications**; your name opens the account menu.
- **Projects** shows a card per project: its cover sheet, name, company, your role, how many sheets, markups, open items and people, and the last activity. **Search projects** by number or name; sort by Name or Number. Click a card to open it; its ⋯ (or right-click) opens the project menu.
- **On this PC** shows how much room the Crew drawings take, with the download settings 12.7.
- **Account**: **People**, **New company…**, **Sign out**, and **Connect Procore**.

Start a project



New project: where the drawings come from.

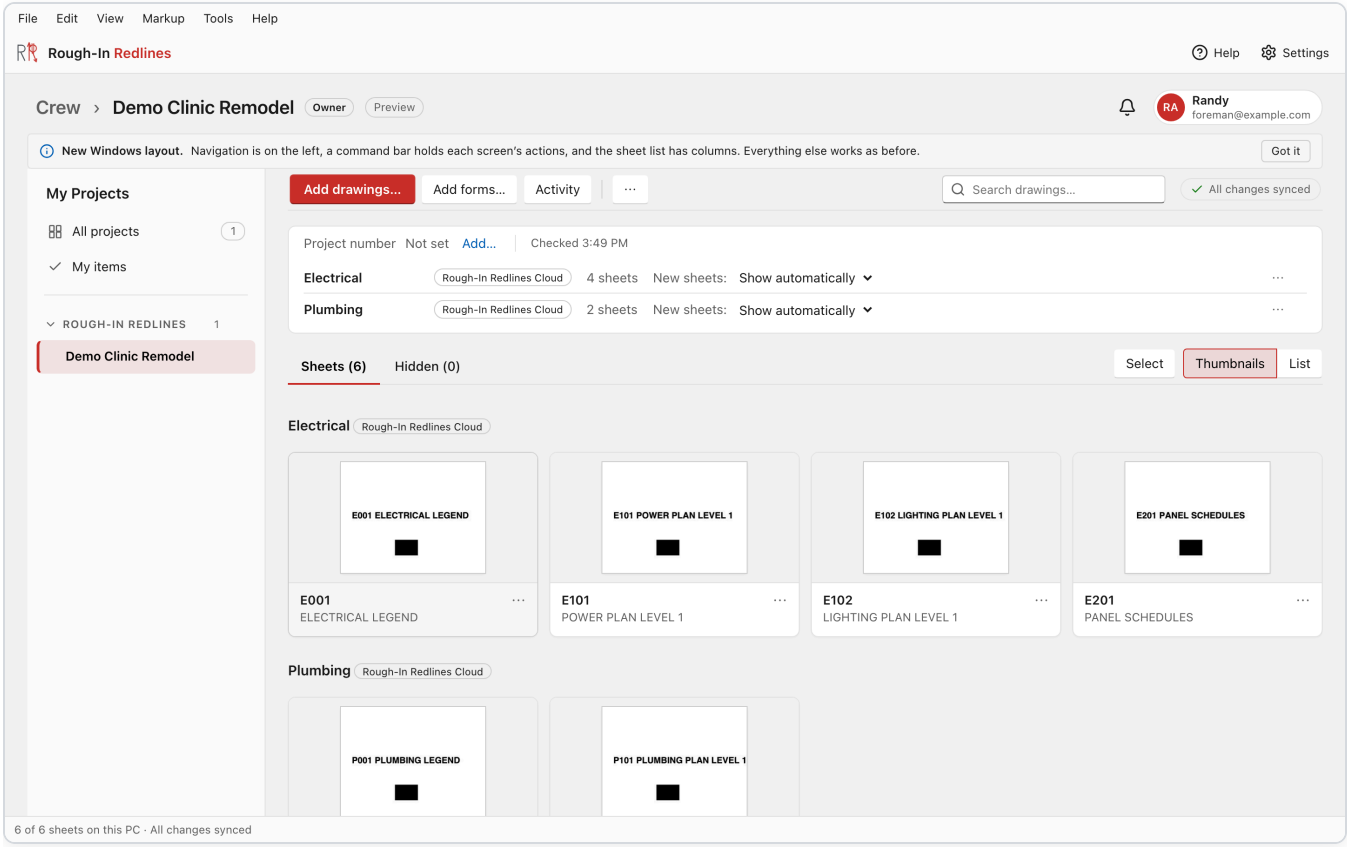


A Rough-In Redlines Cloud project.

1. Click **New project**. If you're in more than one company, pick the company (its people, plan and share links are that company's).
2. Choose where the drawings come from: **Procore** (link a Procore project; drawings come from Procore with your own login, see 12.12) or **Rough-In Redlines Cloud** (upload the set from this PC). You can add sets of either kind later.
3. For a Cloud project, type the **Name** and the **Number** (job number, optional) and click **Create**. Then pick the PDFs to upload (12.6).

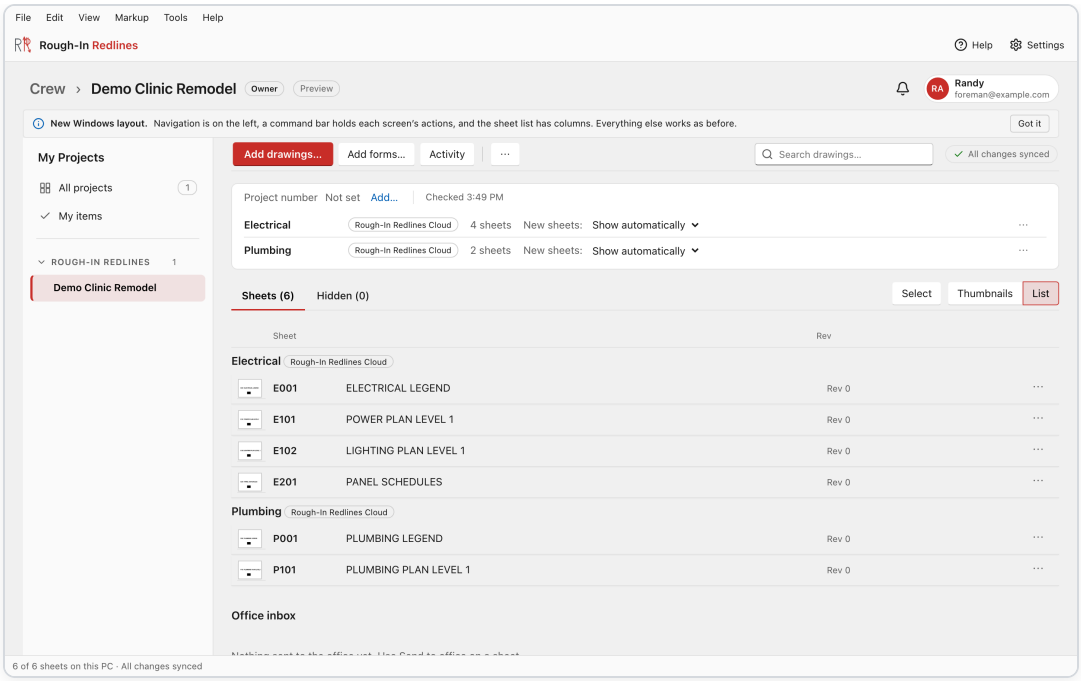
The project number shows before the name everywhere ("24-117 · Name"), on punch lists, notes pages and share links. Managers can change it from the project screen, and **Rename…** it from the project menu.

12.5 Inside a project



A project with two drawing sets, shown as thumbnails.

- The command bar: **Add drawings…**, **Add forms…**, **Activity**, ⋯, the **Search drawings…** box (sheet numbers, titles and the text on the sheets; `Ctrl` + `F`) and the sync status: **All changes synced**, **Syncing**, **Offline**, or **Couldn't sync** with **Retry**.
- The **drawing sets**, each with its source and how many sheets. **New sheets:** chooses whether new sheets **Show automatically** or go in **Hidden**. A set's ⋯ has **Choose sheets…** and **Add drawings to this set…**.
- **Documents**: the forms added with Add forms….
- The tabs **Sheets** and **Hidden**, **Select**, and **Thumbnails** or **List** view. Each sheet shows its number and title, a ✎ when it has markups, its revision, open items, **Not sent yet** (markups not on Crew yet), changes since you last looked, and whether it's on this PC (**Click to download**, **Downloading…**).
- **Since you last looked**: a banner of other people's changes ("Mike added 4 notes on E101 and E103") with **Show on sheets** and **Dismiss**.
- The **Office inbox** at the bottom: sheets sent to the office.



The same project in List view.

A sheet's ⋯ menu (or right-click): **Open** · **Share…** · **Send to Procore…** · **Remove from this PC** / **Download to this PC** · **Compare with** the previous revision · **Replaces…** · **Edit number / title…** · **Send to office…** · **Add notes page…** · **Download notes (PDF)…** · **Hide** / **Hide for everyone**.

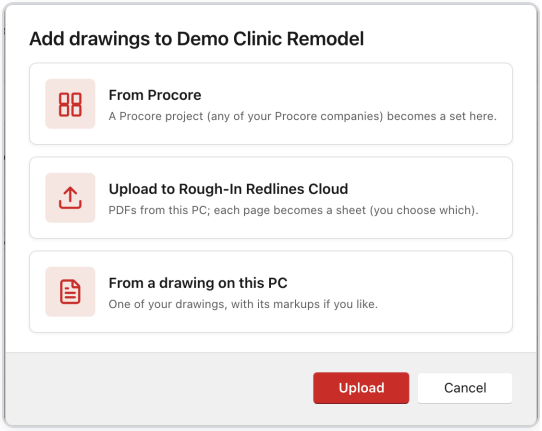
The project's ⋯ menu: **Items** · **People on this project** · **Rename…** · **Share files…** · **Check for new drawings** · **Download all notes (PDF)…** · **Archive project…** · **Remove from my list** · **Delete permanently…** (archived projects only).

12.6 Add drawings and forms

Click **Add drawings…** and pick where they come from: **From Procore**, **Upload to Rough-In Redlines Cloud** (PDFs from this PC; each page becomes a sheet, you choose which), or **From a drawing on this PC** (one of your drawings, with its markups if you like). Or just drop PDFs on the project.

Before anything is uploaded, you check the sheets:

1. **Set**: **New set named …** or **Add to** an existing set.
2. **Issue**: name it (*Rev 0, IFC, ASI 3…*).
3. Check each row's **Sheet** number and **Title**, suggested from the bookmark, page label, title block or file name. The **Status** says **NEW**, **CHANGED** or **UNCHANGED**. Unchanged sheets aren't added again, so everyone downloads only what changed. A number that's already in the project becomes **a new revision**.



Add drawings to a project.

4. Click **Upload N sheets**. The upload carries on in the background, and picks up again after a restart.

- A PDF with fillable fields asks whether to add it as a form or as drawings. **Add forms…** adds each PDF as one document (all pages, fields fillable) under **Documents**.
- **Replaces…**: when a sheet was renumbered in a new issue, pick the old sheet it replaces; **Carry the markups?** → **Carry** brings your markups across, and the old sheet goes to Hidden.
- **Share to Crew…**: from a drawing on your Home tab (right-click → Share to Crew…, or **File** → **Share to Crew…**): pick the company and project (or **New Cloud project…**), the set name, and tick **Bring my markups along (as my layer)**. The drawing on this PC stays as it is.

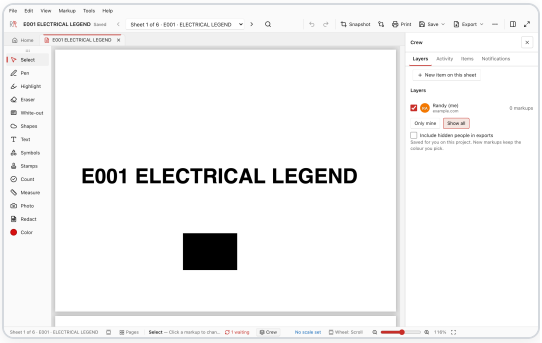
12.7 Offline, downloads and storage

The sheets you chose download by themselves, so they open without internet. When a set is first linked you're asked which sheets you want (by discipline); the rest go to Hidden. Markups you make offline are kept and sync when you're back online ("Offline — changes sync when you're back online").

- **Remove from this PC** frees the room a sheet takes; the sheet stays in the list and downloads again with a click. Your markups stay.
- On the Crew home screen, under **On this PC**: **Download only on Wi-Fi or wired networks (not on metered connections)** (iPad: **Download on Wi-Fi only**), and the **Storage limit**: 2, 5, 10 or 20 GB, or No limit. When the limit is reached, old revisions are removed first.
- The office inbox and the markup history need a connection.
- IPAD The iPad also checks for new revisions in the background and keeps downloading.

12.8 Crew sheets in the editor

A Crew sheet opens like any drawing, with every tool. The whole set scrolls in the same tab, in the project's order; the status bar says which sheet is in view ("Sheet 3 of 27 · E101 · Lighting Plan"). Only the sheet of the tab is edited: click another sheet to work on it.



The Crew panel's Layers tab.

The **Crew** button in the status bar (iPad: the Crew menu in the top bar) opens **Layers**, **Activity**, **Items** and **Notifications**:

- **Layers**: one row per person, in their color, with how many markups. Tick or untick people, or **Only mine** / **Show all**. **Include hidden people in exports** decides whether hidden layers go into exports.
- Each markup's history, with **Restore deleted**. When a teammate changed your edit: **Restore mine**.

Send to office… (sheet menu, or More in the editor) sends the sheet to the project's office inbox as a PDF with your markups burned in, with an optional note. It works offline: it goes when you're back online.

12.9 Items, comments and notifications

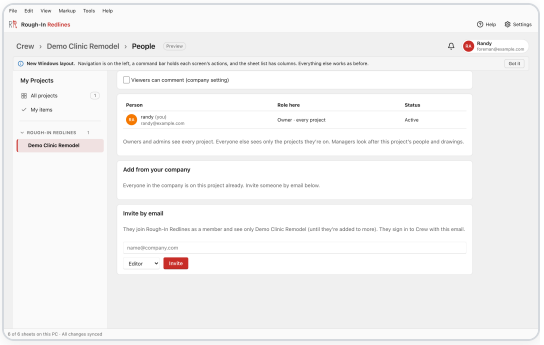
- **Items** are a punch list: right-click a markup → **Assign…**, or right-click empty sheet → **New item on this sheet…**. Fill in **What needs doing**, **Assign to**, **Due**, **Priority** (Low, Normal, High, Urgent), **Grid location** and **Details**. Items go from Open, In progress and Done to Verified (or Reopened). **My items** lists yours across projects, with filters, and **Export PDF** or **Export Excel (CSV)**. Only editors and managers can be assigned items.
- **Comments**: right-click a markup → **Comments…**; type @ to mention someone.
- **Notifications** (the bell): items assigned to you, comments, @mentions and new revisions. **Settings** there chooses, per kind, in the app, push, and email (**Right away**, **Daily digest** or **Off**). Push goes to your iPads; Windows shows notifications while the app is open.

12.10 People

People on this project (project ⋯ menu, managers) lists each person, their role here and status. **Add from your company** picks a person and a role; **Invite by email** invites someone new as a member who sees only this project. **Remove from project** stops them seeing it right away; their markups stay.

People in the company (Account → People; owners and admins) has **Invite someone**: their email, a role (Member, Admin or Owner) and, for a member, the projects they're on. They get an email, open Crew on their iPad or PC and sign in with that address. The list shows whether each person is **Active** or **Invited — hasn't signed in yet**, with **Remove**.

Owners can also set **Company name…** and **Email domain…** (**Use @yourcompany.com**: anyone signing in with a work email there becomes a member, with no projects until someone adds them).



People on this project.

12.11 Share links

Send plans, specs and any other files with a link: no OneDrive needed, and the person doesn't need the app. Use **Share files…** (project ⋯ menu or the company menu) or a sheet's **Share…**.

1. Drop files or **Add files…** / **Add folder…** (PDFs, Word, Excel, any type, up to 200 MB each), or add sheets **From a project**. Add a **Title** and **Note** if you like.
2. **Link works for**: 1 day, 7 days, 30 days or **On a date…** (up to a year). **Downloads**: Unlimited, One-time, or Up to….
3. Tick **Needs a PIN** (4–6 digits) for sensitive files, and send the PIN separately, by text or a call.
4. Click **Create link**, then **Copy link**. A link is shown only once, when it's made.

Shared links lists your links (owners and admins see everyone's) with **Revoke** and **Extend**. Share links are download links, not live views: files are removed 30 days after a link expires or is revoked. Owners and admins can brand the download page with the company name and logo (**Share link branding**).

12.12 Procore

Settings → **Procore** turns Procore on or off. It is off until you turn it on. Off hides Add from Procore, Connect Procore and Send to Procore; projects that already have Procore sets still open their sheets.

1. **Connect Procore** (Crew home → Account, or the account menu). Sign in to Procore in your browser, then come back to the app. Each person uses their own Procore login.
2. **New project** → **Procore** → **Use Procore**: **Pick a Procore project** from any of your Procore companies, name it and click **Create project**. Then choose the sheets you want.
3. New drawings published in Procore come in with **Check for new drawings**; **Add drawings…** → **From Procore** adds more sets.

Send to Procore… (editors and managers) sends a sheet with your markups to the project's Documents folder (Field Redlines), a new Observation, a new RFI (draft) or as a photo. **Publish to Procore…** (managers) publishes the sheet with your markups as a new revision in Procore's Drawings, for a Drawings admin to review. **Unlink from Procore…** keeps the sheets as a Cloud copy or removes them.

> **"ASK YOUR PROCORE ADMIN…"**
> If Procore says the app isn't installed for your company, ask your Procore admin to install it (Procore → Company Admin → App Management → Install Custom App), or click **Upload instead**.

12.13 Archive, remove and delete projects

- **Archive project…** (managers, owners and admins): it leaves everyone's list and becomes read-only; nothing is deleted. Its files are freed from your PC. Restore it from **Archived projects**.
- **Remove from my list** (anyone): hides it for you only. It comes back if someone mentions you or assigns you an item; **Show again** from **Hidden projects**.
- **Delete permanently…** (owners and admins, archived projects only): deletes the project for everyone, with its sheets, markups, notes, items, comments, files and share links. It can't be undone. Download the notes and items first, then type the project name to confirm and click **Delete**.

12.14 Sign out

Account → **Sign out…**. Your Crew projects, their sheets and your Procore connection leave this device, so the next person can sign in with their own email. Your own drawings (not in Crew) stay. Changes not yet sent are sent first; if you're offline, sheets with unsent changes are kept as copies under Home → **Saved copies** ("E101 (unsynced, Project)"), and the button says **Sign out anyway**.

13 Troubleshooting and questions

Answers to the questions people ask most, and what to do when something doesn't work.

13.1 Troubleshooting

A tool or menu item is missing

It is probably switched off in **Settings** → **Tools**. A help topic for a tool that's off says so: "… is turned off on this PC — Settings → Tools turns it back on." On a narrow window, the last tools are under the **…** button at the end of the toolbar.

My first click didn't draw anything

A sheet always opens with **Select**, so the first click never draws by accident. Pick the tool first. IPAD If your finger pans instead of drawing, turn on **Finger draws**; it turns off by itself when an Apple Pencil is used.

"Couldn't save" or "Not saved!"

WINDOWS The drawing couldn't be stored on this PC, most often because the disk is full. Click **Not saved — Retry** in the status bar after freeing some space, or use **Save** → **.redline file** to keep a copy somewhere else (another folder or a USB drive). When you close a drawing that isn't saved, the app asks **Save a copy…** or **Close without saving**.

A drawing was "put back from this PC's backup"

Part of the app's storage couldn't be read, so the drawing was restored from the backup the app keeps on the PC. Check that your latest changes are there. If the Home tab says a drawing isn't in the backup either, open its .redline or PDF file again.

Lengths look wrong

- Check the scale in the status bar. Each sheet has its own scale.
- Was the sheet printed at another size (half size, fit to paper)? Calibrate on a known dimension or the graphic scale bar.
- A detail drawn at another scale needs its own calibration while you measure it.
- Photos only measure true when taken straight down.

Find & Count finds too many or too few

Slide **Sensitivity** toward Strict or Loose; box a tighter symbol on the plan (not in the legend); tick **Also other sizes**. "No linework in that box" means the box was on empty paper or text only. On scanned sheets, count by hand.

Find text finds nothing

The sheets are probably scans: pictures with no text in them. Ask the office for a PDF exported from CAD or Revit.

Markups are missing from my export

A kind of markup is hidden in the **Markups** tab (Show), or, on a Crew sheet, a person's layer is hidden. Hidden markups are left out of exports and prints. Show them again, or on a Crew sheet tick **Include hidden people in exports**.

I can't rearrange the pages

The PDF is protected (encrypted). Save an unprotected copy first.

The print isn't to scale

Pick **Actual size (100%)** or an exact reduction such as **Half size (50%)**, not Fit to paper, and keep the printer's own scaling at 100%.

My plotter doesn't show up on the iPad

Many older plotters don't support AirPrint. Send the Flattened PDF to the plotter maker's app (HP Smart / HP Click, Canon PRINT), or print from the Windows app.

Crew: sheets won't download

- *Downloads resume with a connection*: you're offline.
- *Downloads resume on Wi-Fi or a wired network*: **Download only on Wi-Fi** is on and you're on a metered connection.
- *Storage full*: raise the **Storage limit** on the Crew home screen, or **Remove from this PC** sheets you don't need.

Crew: my changes say "Not sent yet" or "Couldn't sync"

They're kept on your device. Click **Retry**, or wait until you're back online. "Waiting for a Crew subscription" means the company's plan needs attention; your changes are kept until it's sorted.

License key messages WINDOWS

- *That license key wasn't found*: copy and paste it from the purchase email.
- *This key is already active on the maximum number of PCs*: Deactivate this PC on the PC you no longer use.
- *This license key has been disabled (for example after a refund)*: contact support if that looks wrong.
- *Your license hasn't been checked for N days*: connect to the internet and click **Check now**.

13.2 Frequently asked questions

Does it work offline?

Yes, every tool. Only sending files, backups, Crew sync and the one-time Pro activation need a connection.

Will my markups open in Bluebeam?

Yes. Export the **PDF for Bluebeam / Acrobat**: each markup arrives as a real PDF markup with its author, subject and date, and measurements and counts arrive as Bluebeam measurements. Or send a flattened PDF for a read-only copy.

Can I use my Bluebeam tool chest?

Yes, on Windows: **Tools** → **Palettes** → **Import a palette…** and pick the .btx file. Each stamp becomes a captured symbol in a new palette.

Is the original PDF changed?

No. Each drawing is copied into the app; the file you opened is never touched.

How do I get a drawing from my PC to my iPad?

Save it as a .redline file and send it (email, OneDrive, Teams…); tap it on the iPad. See 11.8.

Can I count devices and get the totals in Excel?

Yes: Count or Find & Count, then **Export** → **Counts (CSV)**.

Can I fill in and sign a PDF form?

Yes: open it, click **Fill form**, type, tick, sign, then export or print. See 9.1.

Is my signature a digital signature?

It's an electronic signature, a picture of your signature like Acrobat's Fill & Sign, not a certified digital signature. It stays on your device.

Do I need Procore for Crew?

No. Upload the drawings to Rough-In Redlines Cloud instead, or use both.

Is there a subscription?

Not for the app: Rough-In Pro is a one-time purchase, and the free version covers 3 drawings. Only Crew is a monthly plan, and viewers are free.

Can I use one Pro purchase on Windows and iPad?

No, Pro is bought separately in each store. A Windows key works on 2 PCs; an App Store purchase restores on your other Apple devices; a Google Play purchase on your other Android tablets.

Where do I get help?

Press `F1` (iPad: **How to use**) for the built-in help. For anything else, email **support@roughinredlines.com** with your device, what you were doing and a screenshot, or see roughinredlines.com.

14 Shortcuts and gestures

Every shortcut is also shown next to its menu item. On Windows, **Help → Keyboard shortcuts** (`Ctrl` + `/`) lists them all. Single-letter keys work when you're not typing in a box.

14.1 Keyboard shortcuts WINDOWS

Tools

`V`	Select
`P`	Pen
`H`	Highlight
`E`	Eraser
`W`	White-out
`R`	Shapes
`T`	Text
`S`	Symbols
`K`	Stamps
`C`	Count
`M`	Measure
`D`	Redact
`G`	Snapshot
`I`	Fill form
`L`	Symbols: Connect on / off

File

`Ctrl` + `Shift` + `H`	Home
`Ctrl` + `O`	Open…
`Ctrl` + `N`	Blank sheet…
`Ctrl` + `W`	Close
`Ctrl` + `S`	Save .redline
`Ctrl` + `Shift` + `S`	Save as…
`Ctrl` + `P`	Print…
`Alt` + `F4`	Exit

Edit

`Ctrl` + `Z`	Undo
`Ctrl` + `Y` / `Ctrl` + `Shift` + `Z`	Redo
`Ctrl` + `X` / `C` / `V`	Cut / Copy / Paste
`Ctrl` + `D`	Duplicate
`Del` / `Backspace`	Delete
`Ctrl` + `A`	Select all
`Ctrl` + `F`	Find…
`F3` / `Shift` + `F3`	Find next / previous

View

`Ctrl` + `+` / `Ctrl` + `-`	Zoom in / out
`Ctrl` + `0`	Fit to window
`Ctrl` + `1`	Actual size (100%)
`PgDn` / `PgUp`	Next / previous sheet (continuous scroll: a window down / up)
`Home` / `End`	First / last sheet
`Ctrl` + `Shift` + `P`	Pages…
`Ctrl` + `Tab` / `Ctrl` + `Shift` + `Tab`	Next / previous tab
`\`	Side panel
`F`	Focus (hide the tools)
`F11`	Full screen

Markup and Tools

`Ctrl` + `G`	Group (nothing to group: Go to sheet…)
`Ctrl` + `Shift` + `G`	Ungroup
`Ctrl` + `Shift` + `L` / `E` / `R`	Align left / center / right
`Ctrl` + `Shift` + `T` / `M` / `B`	Align top / middle / bottom
`Ctrl` + `Shift` + `F`	Find & Count…
`Ctrl` + `,`	Settings…
`F1`	How to use
`Ctrl` + `/`	Keyboard shortcuts

Mouse and other keys

`Space` + drag	Pan (or drag with the middle or right button)
Wheel / `Ctrl` + wheel	Continuous scroll: scroll through the sheets / zoom at the pointer (`Shift`+wheel scrolls sideways). One page at a time: the other way round. A touchpad pinch always zooms.
`Shift` + drag	Select several (Lasso)
Arrow keys	Nudge the selection (`Shift`: further)
`Ctrl` + `V` with a picture copied	Paste it on the sheet as a picture
Double-click or `Alt`+click a grouped markup	Select that one member to edit it (`Esc`: the whole group again)
`Alt` while placing	Place a symbol freely (no row snap); place a measuring point freely (no snap)
`Esc`	Cancel, clear the selection, then back to Select
`Enter` / double-click	Finish a run; double-click a markup to edit it
Middle-click a tab	Close the tab
`Alt` or `F10`	The menu bar (`Alt` + `F` opens File)
Right-click a markup	Its options, at the pointer
`Shift` + `F10` / Menu key	Options for the selected markup



Help → Keyboard shortcuts.

14.2 Pen, touch and Apple Pencil

Do this	IPAD	WINDOWS touch screen · ANDROID
Draw	Apple Pencil (or one finger with **Finger draws** on)	The pen (or a finger with **Pen and finger**)
Pan	One finger (two fingers while a finger draws)	One finger with **Pen only**; a flick keeps it going
Zoom	Pinch	Pinch
Straight line	Hold the Pencil still for half a second at the end	Hold the pen still for half a second at the end
Markup options	Long-press the markup	Right-click (Android: long-press; with the S Pen, hold the side button and tap)
Erase	Eraser tool	A Surface pen's eraser end works with any tool
Pressure	The Pencil draws wider when you press harder (signatures too)	A Surface, Wacom or S Pen draws wider when you press harder
Full screen	Three-finger tap	`F11` / Focus

Do this	IPAD	WINDOWS touch screen · ANDROID
Change sheet	‹ ›, or swipe sideways in One page at a time	‹ › in the command bar, or scroll

The iPad has no tap-to-undo gesture: use the Undo button (or ⌘+Z on a keyboard). With a keyboard, ⌘+P prints, ⌘+F finds, G is Snapshot, ⌘+C / ⌘+V copy and paste markups, and Delete removes the selection. On a Mac, ⌘+= / ⌘+- / ⌘+0 zoom.

15 Glossary

The words this manual and the app use.

Bluebeam / Acrobat PDF
An exported PDF whose markups stay editable as real PDF markups in Bluebeam Revu or Adobe Acrobat.

Calibrate
Setting a sheet's scale by clicking both ends of a known dimension and typing its length.

Capture Symbol
Copying a symbol off the drawing (for example from its legend) into your own palette.

Cloud (revision cloud)
A scalloped outline drawn around a change.

Company profile
Your company name, logo and office addresses, printed on panel schedules.

Compare
Laying a new print over an older one to show what was added (blue) and removed (red).

Connect (run)
Symbols joined one after another by arcs or straight lines, to show circuiting.

Continuous scroll
Every sheet of a set one under the other, scrolled like a long document. The other choice is One page at a time.

Copy & Place
Dropping copies of one markup, at the same size and angle, at every click.

Count / count mark
A named tally (for example "Duplex receptacles"); each numbered check on the sheet is a count mark.

Count page
A page at the end of a flattened PDF or print listing symbol counts by trade and measured runs.

Counts CSV
A spreadsheet file (opens in Excel) with the counts per sheet, trade and symbol.

Crew (Rough-In Crew)
The shared, synced projects of a company, with roles, offline sheets, items and share links.

Rough-In Redlines Cloud
Where drawing sets uploaded to Crew are kept, for everyone on the project.

Drawing
One PDF set, picture or blank sheet in the app, however many sheets it has.

Find & Count
Boxing one printed symbol and letting the app find and count every one like it.

Flattened PDF
An exported PDF with the markups burned in, so they can't be moved or removed.

Focus
Hiding the tools to see more of the drawing (F).

Group
Several markups that are selected, moved and copied as one.

Issue
In Crew, the name of an upload of a set: Rev 0, IFC, ASI 3….

Items
Crew's punch list: things that need doing, assigned to someone, with a due date and status.

Layers
On your own drawings, the kinds of markup you can show or hide (Ink, Text, Symbols…). On a Crew sheet, each person's markups.

Lasso
Selecting several markups by dragging around them (`Shift`-drag on Windows; the Lasso tool on the iPad).

Local project
A folder for one job's drawings, kept on this device only (Projects on this device).

Markups list
The Markups tab: every markup on the sheet, with show / hide by kind.

Notes page
An editable page of field notes added to a drawing.

Office inbox
In a Crew project, the sheets people sent to the office with Send to office.

Palette (My Palettes)
Your own set of symbols. Shared as a .redlinepalette file.

Panel schedule
A panel's circuit directory, made on the iPad and sent as Excel and PDF cards.

Photo pin
A numbered pin on the sheet that opens a photo taken there.

Rough-In Pro
The one-time purchase that unlocks unlimited drawings.

Redact
Removing content from the PDF for good (unlike White-out, which only covers it).

.redline file
One file with a drawing, every markup, the photos and captured symbols; opens fully editable on another PC or iPad.

Revision triangle
A numbered △ placed next to a revision cloud.

Run
A measured line through several points, named and totalled (for example a conduit run). Also a set of connected symbols.

Scale
How lengths on the sheet relate to real ones (1/4" = 1'-0"). Each sheet has its own.

Share link
A download link for files, for people without the app; can expire, need a PIN or allow one download.

Sheet
One page of a drawing set, usually with a number such as E101.

Snap
Points or symbols landing exactly on the drawing's linework, or in line with neighboring markups.

Snapshot
Copying an area of a sheet as a picture to paste into email or Word.

Stamp
AS-BUILT, FIELD VERIFIED and the others: a word with your initials and the date.

Tag
A symbol with a short abbreviation, such as DW or MW.

White-out
A box that covers part of the drawing so you can draw the new work on top. It does not remove what's underneath.

Rough-In Redlines User Manual for Version 2 · RK Technology LLC · Written from the Version 2 apps: Windows Rough-In Redlines Preview, iPad RR Preview and Android Rough-In Redlines 1.0. Features and wording can change in later updates; the built-in help (`F1`, or How to use) always matches your version.